MILITARY SPECIFICATION

CONTROL SET, GYROSCOPE, ATTITUDE-HEADING AN/ASN-129

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, design, test and acceptance requirements for the Control Set, Gyroscope, Attitude-Heading AN/ASN-129, herein designated as Heading Attitude Reference System (HARS) or AN/ASN-129.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation-Packaging, Methods Of
MIL-B-5087	Bonding, Electrical, And Lightning Protection, For Aerospace Systems
MIL-E-5400	Electronic Equipment, Airborne, General Specification For
MIL-C-5541	Chemical Conversion Coatings On Aluminum And Aluminum Alloys
MIL-C-6781	Control Panel, Aircraft Equipment, Rack Or Console Mounted
MIL-P-7788	Panel, Information, Integrally Illuminated
MIL-A-8421	Air Transportability Requirements, General Specification For
MIL-I-8500	Interchangeability And Replaceability Of Component Parts For Aircraft and Missiles
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: OC-ALC/MMEDO TINKER AFB OK 73145, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

MIL-Q-9858	Quality Program Requirements
MIL-M-18012	Markings For Aircrew Station Displays, Design, and Configuration Of
MIL-N-18307	Nomenclature and Identification For Electronic, Aeronautical, And Aeronautical Support Equipment Including Ground Support Equipment
MIL-P-19834	Plate, Identification, Metal Foil, Adhesive Backed
MIL-M-25047	Marking and Exterior Finish Colors For Airplanes, Airplane Parts, and Missiles (Ballistic Missiles Excluded)
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment
MIL-D-26503	Detector, Magnetic Azimuth, And Compensator, Detector, Magnetic Azimuth
MIL-C-26524	Calibration Set, Magnetic Compass, Type A/E37T-10
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification For
MIL-I-27193	Indicator, Attitude ARU-2B/A
MIL-C-38999	Connector, Electrical, Circular, Miniature, High Density, Quick Disconnect, General Specification For
MIL-B-81793	Bearing, Ball, Precision, For Instruments And Rotating Components
MIL-I-83034	Indicator, Horizontal Situtation AQU-6/A
MIL-D-83320	Detector, Magnetic Azimuth DSU-14/A
MIL-C-83488	Coating, Aluminum, ION Vapor Deposited

STANDARDS

FEDERAL

FED-STD-595	Color

MILITARY

DOD-STD-100	Engineering Drawings Practices
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-470	Maintainability Program Requirements (For Systems And Equipment)
MIL-STD-704	Aircraft Electrical Power Characteristics
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures For Packaging And Packing Of
MIL-STD-810	Environmental Test Methods
MIL-STD-882	System Safety Program Requirements
MIL-STD-889	Dissimilar Metals
MIL-STD-965	Parts Control Program

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities
MS17322	Meter, Time Totalizing Miniature, Digital 115 Volt 400 Hz
MS25213	Control Panel, Aircraft Equipment, Typical Installation
MS27651	Meter, Time Totalizing, Hermetically Sealed, 115V, 400 Hz

HANDBOOKS

MILITARY

MIL-HDBK-472	Maintainability Prediction

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issue of the document which is indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

CONTRACTOR SPECIFICATIONS

FAIRCHILD INDUSTRIES

160S417001A	Critical Item Development Specification For Head-Up Display

(Application for copies should be addressed to Fairchild Republic Co, Farmingdale Long Island, NY 11735.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS.

3.1 First article. When specified, sample(s) shall be subjected to first article inspection (see 4.4 and 6.2.e).

3.2 Item definition. The HARS shall sense aircraft roll, pitch, and heading and shall supply one synchro output each in roll, pitch, and heading, one turn rate output, and one validity signal to using equipment in the aircraft.

3.2.1 System interface. The HARS electrical system interconnect requirements are specified in Figure 1.

3.2.2 Components. The HARS shall consist of the following components:

a. Gyroscope, Displacement CN-1466/ASN-129 (FIGURE 2).

b. Base, Gyroscope Mounting, NSN 6615-01-013-9525 (FIGURE 3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

c. Amplifier, Electronic Control AM-6851/ASN-129 (FIGURE 4).

d. Base, ECA Mounting NSN 6615-01-027-9731 (FIGURE 5).

e. Controller, Compass System C-10005/ASN-129 (FIGURE 6).

3.2.3 Compatibility. The HARS shall be compatible with and shall interface with the following associated equipment:

a. Detector, Magnetic Azimuth DT-309 (MIL-D-26503) or DSU-14/A (MIL-D-83320).

b. Indicator, Horizontal Situation, AQU-6/A per MIL-I-83034.

c. Indicator, Attitude ARU-2B/A per MIL-I-27193.

d. A-10 Head-Up Display (HUD) Subsystem per FRC 160S417001.

e. A/E 37T-10 Calibrator Set, Magnetic Compass per MIL-C-26524, also designated MC-1 or MC-1M.

3.3 Characteristics.

3.3.1 Performance.

3.3.1.1 All attitude capability. The HARS shall be capable of providing correct outputs in pitch, roll and heading from every vehicle attitude, independent of the sequence of movement resulting in that attitude, and shall provide gyroscopic reference spin axes that are undisturbed in angular position by any aircraft movement or sequence of movements. The pitch angle shall have a normal range of ±90°. When pitch passes through the pitch zenith, the roll and heading outputs shall be reversed in phase. In wings level inverted flight, the roll output shall be 180°, and the pitch output shall be 0°.

3.3.1.2 Roll performance. The gyroscope assembly roll output shall provide a measure of angular displacement about the longitudinal axis of the aircraft. The assembly roll-output angle shall increase for a rotation about the aircraft longitudinal axis that causes the right wing to drop below the horizontal plane. Roll shall have a range of ±180°. The gyro assembly shall accept input roll rates up to 1,000° per second. The roll output information shall indicate the position to within ±1.0° for a 360° 2-rpm rotation in the gyro axis. The roll output shall follow up smoothly and shall not move in increments larger than ±0.1°.

3.3.1.3 Pitch performance. The gyro assembly pitch output shall provide a measure of angle between the aircraft longitudinal axis and the local horizontal plane. The pitch output angle shall increase for an aircraft movement about the lateral axis that causes the nose of the aircraft to rise above the horizontal plane when the aircraft is in noninverted flight. The gyro assembly shall accept pitch input rates up to 36° per second. The pitch output information shall indicate pitch position to within ±1.0° for a ±360° 2-rpm rotation in the pitch axis. The pitch shall follow up smoothly and shall not move in increments larger than ±0.1°.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:54Z
Check the source to verify that this is the currentversion before use.

3.3.1.4 Heading performance. The heading output shall provide a measure of displacement about the local vertical axis. The gyro assembly heading-output angle shall increase for a clockwise (cw) movement of the aircraft about the vertical axis. The gyro assembly shall accept heading input rates up to 36° per second. Heading shall have a range of 360°. The heading output information shall indicate changes of the displacement gyro assembly position to within ±0.5° for a rotation in heading of ±360° at 2 rpm when the system is in the DG heading mode. The heading output shall follow up smoothly and shall not move in increments larger than ±0.1°. The heading output shall be reversed 180° as the pitch angle passes through the pitch zenith.

3.3.1.5 Erection cycle.

3.3.1.5.1 Vertical gyro erection. The vertical gyro erection mechanism shall incorporate separate systems for pitch and roll. Erection shall be accomplished by acceleration sensors and erection torquers. The erection sensors shall be positioned and aligned on the gimbals in a manner that will isolate their outputs from the pitch and roll displacements of the aircraft; in particular, the output of the roll erection sensor shall not be affected by aircraft pitch attitudes of ±60°. The erection system shall provide both initial and final erection and shall erect the gyro from any attitude to the local vertical as determined by the acceleration sensor.

3.3.1.5.2 Initial vertical gyro erection. Erection in pitch and roll shall be automatically sequenced following application of power. The vertical gyro spin axis shall assume a position of not greater than 0.75° within 2 minutes after power is applied. The gyro rotors shall have been at a standstill with spin axes in any random attitude at the time of power application. After 2.5 minutes, the vertical gyro spin axis shall align with the direction of local vertical within ±0.25°. When the erection control senses a signal equivalent to 0.5° ±0.25°, normal erection shall begin automatically.

3.3.1.6 Erection rates.

3.3.1.6.1 Normal rates. The normal erection about the pitch and roll axis shall be 1.50° ±0.5° per minute.

3.3.1.6.2 Fast erection. The fast erection about the pitch and roll axes shall be 24° ±4° per minute. Means for external manual operation of the fast erection circuits shall be provided.

3.3.1.7 Erection and slaving control.

3.3.1.7.1 Pitch erection cutoff. Pitch erection shall be cut off when a longitudinal acceleration in excess of 0.085 ±0.01g is sensed by the fore and aft acceleration sensor.

3.3.1.7.2 Roll erection cutoff. Roll erection cutoff shall be a function of rate-of-heading change and shall coincide with slaving cutoff. Control shall be provided by the electronic control amplifier. Roll erection cutoff shall take place during normal or fast erection mode when a turn rate exceeding 8° per minute is applied.

3.3.1.7.3 Heading slaving cutoff. The slaving of the directional gyro shall be cutoff when either or both of the following conditions exist:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

a. The turn rate exceeds 8° per minute.

b. The longitudinal acceleration exceeds 0.085 ±0.01g.

3.3.1.8 Warmup. The warmup time shall be a maximum of 2.5 minutes of operation over temperatures ranging from +71° to -10°C, and shall not exceed 5 minutes at -54°C. After 2.5 minutes of operation over temperature range from +71° to -10°C, the gyro shall be erected within ±0.25°. After 5 minutes of operation at -54°C, the gyro shall be erected within ±0.75°.

3.3.1.9 Static accuracy.

3.3.1.9.1 Roll accuracy and threshold. The roll output shall be 0° ±0.25° when the assembly is mounted to a horizontal surface with the mounting base equipment specified herein. Angular movement of ±0.1° in the roll axis about the roll level position shall be detectable in the roll output information.

3.3.1.9.2 Pitch accuracy and threshold. The pitch output shall be 0° ±0.25° when the gyro assembly is mounted to a horizontal surface with the system mounting base equipment specified herein. Angular movement of ±0.1° in the pitch axis about the pitch level position shall be detectable in the pitch output information.

3.3.1.9.3 Heading accuracy and threshold. The static accuracy of the heading output shall be ±0.5°. A movement of the gyro attitude assembly about the vertical axis equal to or greater than ±0.1° shall be detectable in the gyro assembly heading output.

3.3.1.10 Drift.

3.3.1.10.1 Roll and pitch drift (spin axis vertical). The maximum vertical free drift in roll and pitch shall not exceed 6° per hour exclusive of the earth's rate. The maximum earth rate compensated drift in roll and pitch shall not exceed 7° per hour.

3.3.1.10.2 Roll and pitch drift (spin axis deflected). The maximum free drift in roll and pitch exclusive of earth's rate with the spin axis initially deflected 20° east shall not exceed 10° per hour.

3.3.1.10.3 Heading drift. The maximum free drift in heading with the assembly set in the DG mode shall not exceed 1.0° per hour where the earth's rate is removed from the data analytically and drift bias is electrically compensated.

3.3.1.10.4 DG drift bias adjustment. A DG drift bias adjustment shall be provided on the displacement gyroscope to compensate for DG drift. The DG drift bias potentiometer shall be accessible from the outside of the sealed case. The potentiometer shall incorporate a removable cover to prevent inadvertent change in adjustment. The DG bias adjustment shall provide a minimum range of 4.5° per hour when interfaced with an electronic control amplifier.

3.3.1.11 Power interruption.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the currentversion before use.

3.3.1.11.1 Vertical. Means shall be provided in the gyro system to prevent the system from recycling through an initial fast erection when power has been removed for 40 ±10 seconds. When the power has been removed for 20 seconds and the gyro displaced through angles of ±15° in pitch and roll and then returned to its original position, the pitch and bank axes shall indicate the original position to within ±1° within 10 seconds after reapplication of power.

3.3.1.11.2 Heading. Means shall be provided for maintaining the heading output through a complete power dropout of 10 seconds such that the heading ouput, after the power interruption is removed, shall not differ from the heading output prior to power removal by more than 1°.

3.3.1.11.3 Protection. Means shall be provided to prevent damage to the gyro when it is banked 45° and pitched 45° and power is interrupted sufficiently long to make the system recycle. Power interruption shall not exceed 25 seconds after the time required for the system to recycle.

3.3.1.11.4 Power shutdown. Means shall be provided to prevent gyro damage when the gyro case is rotated at a rate of 30° per second in pitch, roll and azimuth, after the power has been removed for 5, 10, and 15 minutes. The gyro case shall be rotated each time for at least 180°.

3.3.2 Physical characteristics.

3.3.2.1 Weight. The total weight of the HARS shall not exceed 27.0 pounds.

3.3.2.2 Volume. The total volume of the HARS shall not exceed 0.44 cubic feet.

3.3.2.3 Component envelopes. The component envelopes and mounting bases are specified in Figures 2, 3, 4, 5, and 6.

3.3.2.4 Electrical power requirements. The power required by the HARS shall be 4 wire 200/115 V, 3 phase 400 Hz, wye connected in accordance with MIL-STD-704; controller lighting shall use 5 Vac. The power dissipated by the HARS shall not exceed the following values:

a. Starting: 120 VA at 0.7 power factor (84 watts).

b. Running: 75 VA at 0.6 power factor (45 watts).

c. Controller Lighting: 5 VA, 5.5 VA.

The reference phase for internally excited rotary transducers in the displacement gyroscope shall be ϕC.

3.3.2.5 Transient suppression. The system shall have the capability to withstand alternating current powerline transients of 600 V with a duration of 50 microseconds, and steady-state voltage of 200 V for 10 seconds without system degradation or maintenance action required as a result of these voltages.

3.3.2.6 Displacement gyroscope assembly. The displacement gyro assembly shall consist of a hermetically sealed gyro subassembly which in turn shall consist of vertical gyro and directional gyro elements mounted in a multiple

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

gimbal set. The gimbal set shall incorporate a redundant roll gimbal to avoid the gimbal-lock position.

3.3.2.6.1 Vertical gyro. The vertical gyro shall have sufficient angular momentum and shall be so designed to meet the performance specified herein.

3.3.2.6.2 Directional gyro. The directional gyro shall have sufficient angular momentum and shall be so designed to meet the performance specified herein.

3.3.2.6.3 Acceleration sensors. Acceleration sensors shall be affixed to the vertical gyro gimbal to provide a measure of lateral and longitudinal acceleration. These sensors shall provide vertical-erection-error signals, and erection and slaving cutoff information.

3.3.2.6.4 Erection torquers. Torque motors shall be affixed to the vertical gimbal set to provide erection precession control for the vertical gyro. These torque motors shall be capable of erection at normal or fast rates as commanded by the erection electronics.

3.3.2.6.5 Output devices. Three transducers shall be affixed to the gimbal set to provide one roll, one pitch, and one azimuth output from the gyro assembly.

3.3.2.6.6 Case. The displacement gyro assembly case shall form an integral unit for mounting. The gyro assembly shall be provided with a quick-disconnect mount which shall require no special tools for mounting or dismounting. The case shall be hermetically sealed.

3.3.2.6.7 Filling medium. The filling medium shall be at least 98 percent pure, free from dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure.

3.3.2.6.8 Sealing. The hermetic seal shall not be dependent upon materials that will be affected by any conditions to which the unit may be subjected. The leak rate shall not exceed 10^{-4} cm^3/s at an absolute pressure of 14.7 lb f/in^2.

3.3.2.6.9 Weight. The weight of the gyro assembly shall not exceed 13.7 pounds.

3.3.2.6.10 Dimensions. The dimensions of the gyro assembly shall be in accordance with Figure 2.

3.3.2.6.11 Mounting. The mounting of the gyro assembly shall be in accordance with Figure 3 and shall be such that the displacement reference will not deflect by more than $\pm 0.1°$ in roll, pitch, or azimuth as a result of accelerations or any vibrations between 0 and 18 Hz with a double amplitude of 0.010 inch. The torsional resonant period shall be greater than 18 Hz.

3.3.2.6.11.1 Mount weight. The weight of the gyro mount shall not exceed 1.3 pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

3.3.2.6.12 Time totalizing meter (TTM). The displacement gyro shall contain a TTM conforming to MS 17322 or MS 27651. The TTM shall be located as shown on Figure 2 and shall be visible without disassembly of the gyro assembly case.

3.3.2.6.13 Connector. The electrical connector shall be in accordance with Figure 1.

3.3.2.7 Electronic control amplifier. The electronic control amplifier shall contain the heading coupler electronics, system power supply, the gyro gimbal servo electronics, the synchro repeater function for heading, the erection for the vertical gyro, malfunction detection electronics, and the rate-of-turn-output. The electronic control amplifier shall contain no mechanical computing devices or other moving mechanism, except for the heading correction servo. No continuously high-speed rotating devices shall be used. The electronic control amplifier shall provide 23.5 ±2V, 400 Hz single-phase ac excitation to the magnetic azimuth detector. The electronics shall be placed on plug-in cards. Test points shall be provided on the external connector as identified on Figure 1. The cards shall be keyed to prevent placing a card in the wrong slot and shall incorporate folding handles to remove the cards. The handles shall be an integral part of the card. No tool of any kind shall be used to remove the cards, exclusive of the electronic amplifier cover which shall require a screwdriver for removal.

3.3.2.7.1 Heading coupler. The electronic control amplifier shall generate a heading synchronizing angle. This angle shall be modified as required by the selected heading modes.

3.3.2.7.2 Heading coupler action in DG mode. When the DG mode is selected on the compass system controller, the synchronizing angle will be modified (a) as commanded by the SET HEADING control on the compass system controller, (b) to correct for the apparent heading drift due to the earth's rotation at the latitude set on the compass system controller, and (c) to correct for mean directional gyro drift.

3.3.2.7.3 Heading coupler action to SLAVED mode. The heading coupler shall provide magnetic slaving action when the SLAVED mode is selected on the compass system controller. The slaving rate shall be 1.5° ±0.5° per minute.

3.3.2.7.4 Magnetic azimuth detector interface. The heading coupler shall interface with the magnetic azimuth detector to obtain a magnetic-heading-angle reference. The heading coupler shall provide filtering of the magnetic heading. The filtering shall be selected to provide maximum reduction of magnetic azimuth detector oscillation effects. The interface with the magnetic azimuth detector shall allow use of conventional electrical compass swinging equipment for compass calibration.

3.3.2.7.5 Magnetic synchronization errors. The error between the directional gyro output and magnetic reference shall be used to obtain synchronization between the system output heading and the reference, magnetic-heading angle.

3.3.2.7.6 Synchronization. The synchronizing angle shall be incremented at the specified slaving rate in a direction that will reduce the synchronizing error to zero.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the currentversion before use.

3.3.2.7.7 Synchronization angle modification. The synchronization angle shall be modified in the SLAVED mode to:

a. Correct for the apparent heading drift due to the earth's rotation as determined by the latitude set on the compass system controller.

b. Correct the mean directional gyro drift by reference to a voltage and polarity transmitted from the displacement gyro.

3.3.2.7.8 Fast synchronization. Circuitry shall be provided to cause synchronization of the azimuth output at a fast rate not less than 30° per second when operating in the SLAVED mode with the SET HEADING switch on the compass controller depressed, or when switching from DG to SLAVED. The system shall be automatically synchronized at the end of the initialization cycle if the system is in the SLAVED mode.

3.3.2.7.9 Compass calibration. When used in conjunction with a Type A/E37T-10 (MIL-C-26524) (formerly known as Type MC-1M) magnetic compass calibrator set, and DSU-14/A (MIL-D-83320) or DT-309/AJN (MIL-D-26503) magnetic azimuth detector, mounted in the aircraft, the system shall meet the compass calibration performance requirements specified herein.

3.3.2.7.10 Compensation accuracy. The compass calibrating system shall be capable of determining and compensating single-cycle and two-cycle errors from the control assemblies and the detector. The compensated heading errors shall be less than 0.5° rms with no heading error exceeding 1.0° over the complete range of 0° to 360°.

3.3.2.7.11 Compensation. Heading compensation adjustments shall be contained in the electronic control amplifier to compensate for one-cycle and two-cycle errors. The compensation adjustment shall be placed on a removable board which may be installed and removed without taking off the top lid of the electronic control amplifier. The plug-in heading compensation board shall be designed in such a manner as to allow transfer to another electronic control amplifier without the loss of the stored compensation information. Removal of the calibrated compensator from the electronic control amplifier and insertion into a different electronic control amplifier shall not affect the system calibration by more than 1°. The compensator compartment shall be adequately sealed to prevent entry of moisture, sand, dust, or salt spray into the amplifier whenever the compensator is removed from the amplifier. The compensator potentiometers shall be of the environmentally sealed type.

3.3.2.7.12 N-S and E-W (one-cycle) adjustments. Two adjustments will be required to compensate for one-cycle errors. The degree of compensation specified herein shall apply for interconnecting lead lengths between the magnetic azimuth detector and the electronic control amplifier of up to 200 feet of number 28 wire. The compensator shall provide two adjustments by means of which it shall be possible to compensate for one cycle errors produced by the heading system, including the magnetic azimuth detector. One adjustment device shall be capable of compensating for the N-S detector signal error and the other adjustment device shall be capable of compensating for the E-W detector signal error. They shall be adjustable independently and shall be designed to have minimum adjustment interference.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

3.3.2.7.13 Adjustment range. With the detector set on 0° heading, the N-S adjustment shall be capable of changing the output heading a minimum of ±8° in normal field. With the detector setting on 90°, the E-W adjustment shall be capable of changing the output heading a minimum of ±8° in normal field. A clockwise (cw) rotation of each adjustment as specified above shall cause an increase in the resultant heading signal.

3.3.2.7.14 N-S and E-W adjustment interference. A full scale rotation of the E-W adjustment screw shall not influence the system heading, including the magnetic azimuth detector output signal, by more than ±1.5° when it is set at 0° heading. When set at the 90° heading, a full scale rotation of the N-S adjustment shall not influence the system heading, including the detector output signal, by more than ±1.5°.

3.3.2.7.15 Two-cycle adjustments and range. A maximum of two adjustments shall be provided for two-cycle compensation. The amplitude and phase adjustments shall have a nominal adjustment range from 2° to 4° and repeatable accuracy of 9 minutes.

3.3.2.7.16 Heading data repeater. The heading synchro data repeater shall not require adjustment when transferred as an assembly.

3.3.2.7.17 Synchro format.

3.3.2.7.17.1 Isolation. Each synchro output shall be electrically isolated from all other system outputs.

3.3.2.7.17.2 Excitation. The heading synchro output shall be externally excited from a 115 V, 400 Hz power source. The pitch and roll synchro shall be internally excited from ØC.

3.3.2.7.17.3 Gradient. Each synchro format output shall have a no-load, line-to-line voltage of 11.8 ±0.7 Vrms with a phase of 9° ±5° (leading) with respect to the excitation phase.

3.3.2.7.17.4 Synchro impedance and index reference.

3.3.2.7.17.4.1 Output impedance (nominal).

a. Pitch and roll transmitters

Z_{ro} = 945 + j 3297 ohms
Z_{rs} = 1014 + j 451 ohms
Z_{so} = 12.3 + j 33.8 ohms
Z_{ss} = 9.7 + j 3.5 ohms

Null voltage 0.045V rms max.

b. Heading transmitter

Z_{ro} = 950 + j 3850 ohms
Z_{so} = 10 + j 36 ohms
Z_{rs} = 1550 + j 420 ohms

Rotor dc resistance = 700 ohms ±20%
Stator line to line dc resistance = 10 ohms ±20%

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:54Z
Check the source to verify that this is the current version before use.

3.3.2.7.17.4.2 Index reference and sensing.

a. The pitch and roll synchros shall be 180° reference. At 0° X to Z is in phase with ØC and increases for pitch up and right wing down.

b. The azimuth synchro shall be 0° reference. At 0° X to Z is out of phase with the reference Ø and increases for right turn.

3.3.2.7.17.5 Scale error. The maximum error introduced by the synchro data repeaters, when loaded with three control transformers (Bendix AY500-5, or equivalent), shall not exceed 21 minutes of arc. The pitch and roll error at 0° shall not exceed 9 minutes of arc.

3.3.2.7.17.6 Scale error during frequency variation. The maximum scale error introduced by the synchro data repeaters shall not exceed 21 minutes of arc when the excitation frequency differs from 400 Hz by 20 ±5 Hz. The pitch and roll error at 0° shall not exceed 14 minutes.

3.3.2.7.18 Roll erection and heading slaving control. A solid-state azimuth rate sensor, with no mechanical computing device, shall be provided to permit control of roll erection and compass slaving during turns. Under standard conditions, roll erection and compass slaving control shall occur for the following azimuth inputs:

a. Roll erection and slaving shall be removed during turns of ±8°, ±30° and ±180° per minute within 30, 11, and 8 seconds respectively and shall be returned within 20 seconds after the rates have ceased. Erection shall remain ON during turns of ±3° per minute or less.

b. Roll erection and slaving shall be removed within 30 seconds when an input of 30° per minute rate of turn is super-imposed on a ±0.8°, 1-second period azimuth oscillation. Erection and slaving shall remain OFF until the 30° per minute rate of turn has ceased.

c. Vertical gyro roll erection and heading slaving shall remain ON when the following azimuth oscillations are imposed:

±0.8° with a 1.0 second period.
±2.0° with an 8.0 seconds period.

3.3.2.7.19 Pitch erection and heading slaving control. An internal linear acceleration sensor on the displacement gyro and associated circuitry shall be provided to permit control of pitch erection and compass slaving during horizontal fore-aft acceleration. Control shall occur at a threshold acceleration of 0.085 ±0.01g directed along the sensitive axis of the accelerometer at which time the pitch erection and compass slaving shall be cut off.

3.3.2.7.20 Vertical gyro manual fast erection. Circuitry shall be provided with external pins to obtain, when externally activated, a pitch and roll fast erection rate of 24° ±4° per minute only during external activation. If during this period the pitch and roll errors drop below 0.5° ±0.25°, the erection in either or both axes shall revert to normal erection. The roll portion of this control shall only be functional when the turn rate is below 8° per minute (i.e., a rate of turn beyond this threshold after fast erect initiation shall result in the control assembly reverting to roll erection cutoff). Pitch erection shall not be affected by the system turn-rate cutout.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

3.3.2.7.21 Earth rate compensation. Earth rate compensation requirements shall be as follows:

a. Directional gyro - The electronic control amplifier shall accept sin (latitude) from the compass system controller and cause the synchronization angle to rotate at a rate of ±15° sin latitude in degrees per hour. The direction of rotation shall be dependent upon the compass system controller hemisphere switch setting.

b. Vertical gyro - The electronic control amplifier shall accept cos (latitude) from the compass system controller and provide an output to the pitch and roll torquers which will result in rates of:

roll: 15 cos (latitude) cos (heading + MAG VAR)° per hour
pitch: -15 cos (latitude) sin (heading + MAG VAR)° per hour

3.3.2.7.22 Coriolis compensation. Means shall be provided within the electronic control amplifier to compensate the magnetic azimuth detector signal for coriolis acceleration. The coriolis compensation shall be fixed for a set aircraft velocity of 300 knots.

3.3.2.7.23 Rate-of-turn function. The electronic control amplifier shall incorporate the necessary electronics to provide derived rate-of-turn output to the turn indicator of the attitude director indicators. The signal source impedance shall be 500 ohms ±10 percent. The heading rate output shall be a direct-current voltage source proportional to the rate of change of heading. When loaded with 500 ohms, the voltage source shall provide 2 ±0.13 milliamper (mA) at 180° per minute to the load.

3.3.2.7.23.1 Threshold. The maximum threshold shall not exceed 3° per minute.

3.3.2.7.23.2 Time constant. The heading rate output shall have a time constant of 1.75 ±0.25 seconds and an undamped natural frequency greater than 5 Hz.

3.3.2.7.23.3 Saturation. The heading rate output shall saturate at a rate exceeding 380° per minute.

3.3.2.7.24 Malfunction monitoring. The electronic control amplifier shall provide warning interlocks with two independent external outputs. An open circuit shall appear between pins 25 and 26 and between pins 27 and 28 of 1J1 when the conditions described in Table 1 occur.

3.3.2.7.25 Magnetic variation (MAG VAR) control. The electronic control amplifier shall implement MAG VAR correction in order to correct the heading input to the earth rate compensation for the vertical gyro.

3.3.2.7.26 Connectors. The electrical connectors for the electronic control amplifier shall be in accordance with Figure 1.

3.3.2.7.27 Weight. The weight of the electronic control amplifier shall not exceed 9.5 pounds.

3.3.2.7.28 Dimensions. The dimensions of the electronic control amplifier shall be in accordance with Figure 4.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:54Z
Check the source to verify that this is the current version before use.

3.3.2.7.29 Mounting. Mounting of the amplifier shall be of the quick-disconnent type and shall be in accordance with Figure 5.

3.3.2.7.29.1 Mount weight. The weight of the electronic control amplifier mount shall not exceed 1.2 pounds.

3.3.2.7.30 TTM. The amplifier shall contain a TTM conforming to MS27651 or MS17322. The TTM shall be located as shown on Figure 4 and shall be visible without disassembly of the case.

3.3.2.8 Compass system controller. The compass system controller shall be designed in accordance with MIL-C-6781, type I. The configuration shall be in accordance with Figure 6. Fasteners shall conform to MIL-F-25173 and plastic plate lighting shall conform to MIL-P-7788, type IV, except as follows:

a. Paragraph entitled "Connector Location" shall not apply.

b. The contrast, stray light, and dial visibility requirements of MIL-L-27160 shall apply. Stray light shall be measured 1 inch in front of the panel.

c. The compass system controller lighting (white, 4.5V) shall meet the color requirements of MIL-L-27160 and the brightness requirements of MIL-P-7788.

3.3.2.8.1 Prime standard. Panel lighting shall be compared to and shall balance in brightness and color with the prime standard. The prime standard shall meet the requirements of 3.2.2.8. Visual comparisons shall be made in a dark room at a distance of 2 ±0.5 feet.

3.3.2.8.2 Function selector switch. A mode selector switch shall be provided for selecting any of the following modes of operation:

a. SLAVED: In the SLAVED mode, the system shall operate as a gyro stabilized magnetic compass.

b. DG: In the DG mode, the system shall provide heading independent of the magnetic detector as a latitude corrected directional gyro.

3.3.2.8.3 Synchronization and heading set. A push-to-sync manual control shall be provided to effect the fast synchronization and heading set. When in the DG mode, the heading set control markings shall indicate in which direction the control is to be turned to provide a decreasing or increasing heading. The mechanism shall, via operator control, provide a continuously variable drive rate from 0° to 30° per second in either direction. The drive rate shall be so damped as to permit accurate selection of the desired heading. The control shall be of the push-to-turn, spring-to-center type.

3.3.2.8.4 Push-to-sync. When in the SLAVED mode and when the heading set control is pushed, the appropriate push-to-sync circuit in the electronic control amplifier shall be activated; the system shall synchronize within 1° in not more than 6 seconds and the warning interlocks shall open.

3.3.2.8.5 Latitude correction control. A manually set control shall be provided to correct the system for the apparent drift of the directional and

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

vertical gyros due to the earth's rotation. The direction of correction for the Northern or Southern Hemisphere shall be selected by a hemisphere switch. The latitude dial shall be marked with latitude from 0° to 90°, graduated in 2° increments, with major divisions and numerals every 10°. The correction shall be applied in both DG and SLAVED modes of operation. The controller correction shall be equivalent to 15° per hour times the sin and cos of the indicated latitude within a tolerance of ±0.1° per hour.

3.3.2.8.6 Synchronizing indicator. A synchronizing indicator, other than signal lights, shall be provided to indicate synchronization to within ±1/4° between compass system output and the detector, or electrical equivalent, when in the SLAVED mode of operation. When the system is not synchronized, the synchronizing indicator shall indicate in which direction the system must be corrected to avoid an ambiguity. The synchronizing indicator lighting and markings shall afford ease of viewing under all ambient conditions.

3.3.2.8.7 MAG VAR control. A three position switch shall be provided which shall allow setting magnetic variation compensation to the approximate local magnetic field. The three settings shall be 0°, +15° and -15°.

3.3.2.8.8 Markings and dimensions.

3.3.2.8.8.1 Markings. The durability of all markings shall be such that they will withstand usage encountered in service. Unless otherwise specified, the style and proportions of numerals and letters shall conform to MIL-M-18012.

3.3.2.8.8.2 Numerals. Each numeral shall be so distinctly marked that no doubt will exist as to which graduation the numeral applies.

3.3.2.8.8.3 Marking dimensions. The dimensions of the markings shall be as follows:

Marking	Height or Length (Inch)	Width of Line or Graduation (Inch)
COMP (Panel title)	0.125 ±0.005	0.015 ±0.005
LAT	0.125 ±0.005	0.015 ±0.005
2° graduations	0.93 ±0.010	0.015 ±0.005
10° graduations	0.125 ±0.010	0.025 ±0.005
+,-,N,S,DG,SLAVE PUSH-TO-SYNC, MAG VAR, +15, and -15,O,SYN IND,HDG	0.125 ±0.005	0.015 ±0.005

3.3.2.8.9 Color. The background of the dial and all other markings not otherwise specified shall be finished in lusterless black, color No. 37038 of FED-STD-595.

3.3.2.8.10 Touch temperature. The maximum stabilized surface temperature for the controls, displays, and knobs shall not exceed 41°C in an environment of 27°C.

3.3.2.8.11 Weight. The weight of the compass system controller shall not exceed 1.0 pound.

3.3.2.8.12 Connector. The electrical connector shall be in accordance with Figure 1.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the currentversion before use.

3.3.2.8.13 Dimensions. The dimensions of the compass system controller shall be in accordance with Figure 6.

3.3.2.8.14 Mounting. The compasss system controller shall be mounted in accordance with MS25213.

3.3.3 Reliabilbity. The HARS shall have a mean time between failure (MTBF) of 750 hours at 90% confidence level.

3.3.3.1 Longevity. The HARS shall have a longevity of 2500 hours before it consistently fails to meet the MTBF index.

3.3.4 Maintainability. The following maintainability requirements shall apply:

a. Minimization of the complexity of maintenance tasks by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items.

b. Rapid and positive recognition of equipment malfunction or marginal performance through the use of intermediate and depot level support equipment. Predicted durations shall be submitted to the Air Force for approval.

c. Rapid and positive identification of the replaceable defective part, assembly or component. Predicted durations shall be submitted to the Air Force for approval.

d. Minimum numbers and types of tests and test equipment (special and standard) needed to perform maintenance.

e. Optimum accessibility in all systems, equipments and components requiring maintenance, inspections, removals and replacements.

f. Isolation of Electronic Control Amplifier (ECA) malfunction to card level when using intermediate level support equipment.

3.3.4.1 Quantitative maintainability requirements.

3.3.4.1.1 Mean time to repair (MTTR). The MTTR defined in MIL-STD-470 shall be as follows:

a. Electronic Control Amplifier: Not more than 3 hours.

b. Controller: Not more than 1.5 hours.

c. Gyro Assembly: Not more than 10.5 hours based on MIL-HDBK-472 prediction.

d. Gyro Assembly: Approximately 35 hours, including, whenever applicable, preparation, hookup, fault isolation and diagnosis, access, repair, functional test of repair, and reassembly to final configuration.

3.3.4.1.2 Maintenance manhour per flight hour (MMH/FH). The MMH/FH at organizational and intermediate level shall not exceed 0.005 and 0.01 hour, respectively.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

3.3.5 Environmental conditions. The HARS or components shall be capable of meeting the requirements specified herein under the following environmental conditions:

a. Temperature-Altitude: Temperature ranging from -54° to +71°C operating and -62° to +85°C nonoperating at altitudes from sea level to 70,000 feet, as applicable.

b. Explosive atmosphere: Equipment shall operate within an explosive atmosphere without causing an explosion.

c. Humidity: Relative humidity of 95 percent at 71°C.

d. Salt fog: Exposure to salt fog for 48 hours.

e. Fungus: Fungus growth as encountered in tropical climates.

f. Dust: Blowing fine sand and dust particles.

g. Rain: Simulated rain at a rate of 5 ±1 inches per hour for 10 minutes (for electronic control amplifier and compass system controller only).

h. Vibration: The HARS shall operate during/after vibration in accordance with the performance requirements of this specification when subjected to the following vibration spectra (see Figure 7).

(1) Sinusoidal Vibration

Frequency	Level
5-20 Hz	.10 in. DA
20-2000 Hz	±2 g

This spectrum shall be applied as specified in 4.6.5.9.1

(2) Random vibration-normal spectra. The equipment shall operate during the application of the Curve A spectrum of Figure 7. The equipment shall operate after the curve B spectrum of Figure 7 is applied. The spectra shall be applied as specified in 4.6.5.9.2.

(3) Random vibration-gunfire spectrum. The compass controller shall operate during the application of the random vibration-gunfire spectrum of Figure 8. The spectrum shall be applied as specified in 4.6.5.9.2 for the compass controller only.

i. Shock: Shock of 15g operating and 30g nonoperating.

j. Acceleration: 13.5g nonoperating (down) and 9g operating (down).

k. Temperature shock: Sudden changes in temperature of surrounding atomosphere ranging from -54°C to +71°C.

l. Acoustical noise: Acoustical noise of 150 dB for 30 minutes.

3.3.5.1 Insulation resistance. When 500 VDC is applied between the case and the specified pins, as applicable, the minimum resistance shall be 50 megohms.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the currentversion before use.

3.3.6 Transportability. Criteria and limitations for maximum dimensions and weight, as crated for shipment as well as material handling and transportation modes, shall comply with AFR 80-18, except that air transportability requirements shall conform to MIL-A-8421.

3.3.6.1 Transportation requirements. There are no unique transportation or handling requirements. Standard available transportation and handling equipment can be used in movement of this configuration item.

3.4 Design and construction. The design and construction of the HARS shall conform to the applicable requirements of MIL-E-5400 except as modified herein. The HARS shall be designed to accept and provide input and output signals as specified herein and shall be constructed to withstand the strains, jars, vibrations, and other conditions incident to shipment, storage, installation and service. Corresponding components of all systems procured shall be interchangeable between systems of the same type. No gimbal binding under differential pressure conditions shall occur as a result of interference between case and gimbal should a leak develop in service.

3.4.1 Materials, processes and parts. Materials, processes and parts shall be in accordance with the requirements of MIL-E-5400 except as modified herein.

3.4.1.1 Electrical connectors. Electrical connectors shall meet the requirements of MIL-E-5400 except as specified herein. Hand-mated circular electrical connector shall be in accordance with MIL-C-38999.

3.4.1.2 Parts. Part selection shall be within the categories defined in MIL-STD-965.

3.4.1.3 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except:

a. Dials, small holes, and case inserts need not be anodized.

b. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

c. Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

d. Castings containing nonaluminum integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

e. When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

f. Parts in hermetically sealed containers need not be anodized.

3.4.1.4 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used.

3.4.1.5 Fungusproof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. If used within a hermetically sealed enclosure, fungicidal treatment shall not be required.

3.4.1.6 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to conditions likely to be encountered in storage or normal service.

3.4.1.6.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.1.7 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the control assembly except where magnetic materials are essential.

3.4.1.8 Corrosive fumes. Materials, as installed and used in the control assembly and under the service conditions specified herein, shall not liberate deleterious fumes.

3.4.1.9 Protective treatments. When materials are used in the construction of the control assembly that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. Protective coatings that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall not be used.

3.4.1.10 Paint. The case of the displacement gyro and amplifier shall be gray, color No. 36231, and the case of the compass controller shall be black, color No. 37038, in accordance with FED-STD-595.

3.4.1.11 Screws and fasteners. Screws used to attach the system to the aircraft shall be size No. 10 or larger unless specified otherwise herein. Mounting provisions shall be designed so that no mounting screw threads are in bearing. The use of inserts is considered to be threads in bearing. All screws and fasteners having cross recess type heads shall be of the Phillips recessed type. Access covers that must be removed for the organizational or intermediate level maintenance of the equipment shall be captive and shall be secured by a minimum number of captive fasteners.

3.4.1.12 Plug-in cards. The electronics in the electronic control amplifier shall be placed on plug-in cards which shall be removable without the use of tools, exclusive of the electronic control amplifier cover and power supply transformer. The plug-in cards shall be keyed to prevent them from being placed in the wrong slots and shall be color coded or numbered.

3.4.1.13 Cases. Cases shall be in accordance with the specified dimensions and shall provide hermetically sealed enclosures when required for internal

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

mechanisms. The cases shall be so designed that the internal mechanism may be removed, replaced, and the case resealed without the use of special tools unless they are approved by the procuring activity.

3.4.1.14 Bearings. All bearings used in the construction of the HARS shall be in accordance with MIL-B-81793.

3.4.1.15 Soldering. Soldering shall be accomplished in accordance with MIL-STD-454, requirement 5.

3.4.2 Electromagnetic interference and compatibility. The HARS shall be designed to meet the requirements of MIL-STD-461 for Class A-1 equipment.

3.4.2.1 HARS EMI. The HARS and its individual units shall meet the acceptance criteria of MIL-STD-461 for class A-1 equipment when tested in accordance with MIL-STD-462. Specific tests shall be performed in accordance with 4.6.5.10.

3.4.2.2. Electrical bonding and grounding. The electrical bonding and grounding requirements for the HARS shall be in accordance with MIL-B-5087.

3.4.3 Nameplates and product marking.

3.4.3.1 Nameplates. Nameplates shall meet the requirements as specified herein, and shall be in accordance with MIL-N-18307 and MIL-M-25047. Nameplates conforming to MIL-N-18307 shall be attached to the item by screws. Metal foil, adhesive backed nameplates conforming to MIL-P-19834 shall not be used where the item may be subjected to temperature extremes beyond -67 to +221°F. Plastic nameplates shall not be used.

3.4.3.2 Marking. Identification marking shall be applied to the items in accordance with MIL-STD-130. The legend, CAUTION - DELICATE INSTURMENT - HANDLE WITH CARE, shall be stenciled on the gyro case.

3.4.3.2.1 Serialization. The components of the HARS shall be serialized with the last four digits numeric. The serial number shall be included on the nameplates specified in 3.3.3.1.

3.4.4 Workmanship. The HARS shall be manufactured in accordance with MIL-STD-454.

3.4.4.1 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.4.4.2 Cleaning. The control assembly components shall be thoroughly cleaned, and loose, spattered or excess solder, metal chips and other foreign material removed during and after final assembly.

3.4.4.3 Dimensions and tolerances. Dimensions and tolerances not specified shall be consistent with producibility. Where dimensions and tolerances may affect interchangeability, operation or performance of the control assembly, components, or detail part therein, they shall be held or limited for function while preserving producibility.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

3.4.4.4 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage to the screw, bolt threads, or associated components.

3.4.5 Interchangeability. Interchangeability requirements shall be in accordance with MIL-I-8500 and as specified herein. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DoD-STD-100 shall govern the manufacturer's part number and changes thereto.

3.4.6 Safety.

3.4.6.1 Personnel safety. Provisions for personnel safety shall be in accordance with MIL-STD-454, requirement 1.

3.4.6.2 Safety. Selection of appropriate design features to eliminate catastrophic hazards and minimize critical hazards shall be given precedence over corrective or protective features, and shall be in accordance with MIL-STD-882.

3.4.7 Human performance. Design of the HARS shall comply with the human performance requirements and human engineering design standards set forth in MIL-STD-1472.

3.5 Maintenance.

3.5.1 Pre-flight test. Pre-flight testing shall be accomplished without the use of internal or external test equipment by such actions as operation of controls and observing cockpit indicators.

3.5.2 Organization level maintenance. Organization level maintenance shall be accomplished without the use of internal or external test equipment. Organization level maintenance consists or removal and replacement of line replaceable units (LRU's) from the aircraft for operational verifications.

3.5.3 Intermediate level maintenance. Intermediate level maintenance shall consist of verification of faulty LRU's and identification and replacement of faulty plug-in subassemblies.

3.5.4 Depot level maintenance. The following will require depot level maintenance:

a. Displacement gyro.

b. Electronic control amplifier.

c. Compass system controller.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. A quality assurance program shall be established in accordance with MIL-Q-9858. This section describes the tests/verifications required to assure the contractor's compliance with the requirements specified in Section 3. The tests/verifications may be performed at the contractor's facilities, subcontractor's facilities, or any commercial laboratory acceptable to the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

4.2 Classification of tests. The inspection and testing of the Heading Attitude Reference Systems shall be classified as:

a. First article tests.

b. Quality conformance tests.

4.3 Test conditions.

4.3.1 Standard conditions. Unless otherwise specified herein, the following standard test conditions shall be used:

a. Temperature - Room temperature 25° ±10°C.

b. Altitude - Normal ground.

c. Vibration - None.

d. Humidity - Room ambient up to 95 percent relative humidity.

e. Attitude - Level.

f. Input voltage - 3-phase, wye-connected, 400 ±1 Hz, 115 ±2V to common neutral, phase rotation A, B, C.

g. Scorsby motion - Sinusoidal motion (±1.5° excursions at 5 to 7 cpm reversing direction once each minute) roll, pitch, and yaw axes applied to the displacement gyro.

h. Earth's magnetic field - Standard field strength: 14.32 ampere-turns/meter horizontal, 42.97 ampere-turns/meter vertical; weak field strength: 6.37 ampere-turns/meter horizontal, 42.97 ampere-turns/meter vertical; and, strong field: 33.42 ampere-turns/meter horizontal, 42.97 ampere-turns/meter vertical.

4.3.2 Standard test equipment. The following test equipment shall be used.

4.3.2.1 Scorsby table. The Scorsby table shall be capable of making an excursion about each axis of ±15° at 5 to 7 cpm with motion reversal once each minute.

4.3.2.2 Rotary table. The rotary table shall have a 0° to 360° range to test for heading variation of the gyro.

4.3.2.3 MAD simulator. The MAD simulator shall be Sperry part No. 2586546, or equivalent. Headings from 0° to 360° in 15° increments either clockwise (cw) or counterclockwise (ccw) shall be selected to within ±13 minutes repeatable to within 1 minute. Headings shall be provided at both standard and weak field strengths.

4.3.2.4 Helmholtz cage. The Helmholtz cage shall be capable of varying the horizontal field strength at the detector position from 0.06 to 0.42 oersted. The vertical field at this position shall be variable from 0 to 0.54 oersted. Both horizontal and vertical fields shall be continuously variable and shall be independent of each other.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

4.3.2.5 Tilt table. The tilt table shall be so designed that it can be rotated and indexed ±360° in roll, pitch, and heading.

4.3.2.6 Motor-driven simulated gyro transmitter. The motor-driven simulated gyro transmitter shall be so designed that it can be driven at 0 to 50 rpm ±10 percent in addition to the following:

a. Steady state: 60° per second ±10 percent; 360° per minute ±10 percent; 180° per minute ±10 percent; 90° per minute ±10 percent; 45° per minute ±10 percent; 30° per minute ±10 percent; 8° per minute ±3 percent; and 3° per minute ±3 percent. The 3°, 8°, 30°, 45°, 90°, 180°, and 360° per minute rates shall be reversible.

b. Oscillatory (sinusoidal): ±2.0° with an 8-second period and ±0.8° with a 1-second period.

c. Superimposed steady state and oscillatory: 30° per minute superimposed on a ±0.8°, 1-second period oscillation.

4.4 First article testing. First article testing shall consist of all tests specified herein.

4.4.1 Test samples. The test samples shall consist of 7 complete HARS representative of the production equipment. Three shall be subjected to all tests except reliability and the remaining 4 HARS shall be subjected to the LRU burn-in as per 4.6.1.2 and 4.6.2.2, and to the reliability test as per 4.4.2. The samples shall be identified with the manufacturer's serial number, part number and such other information as required by the procuring activity.

4.4.2 Reliability tests.

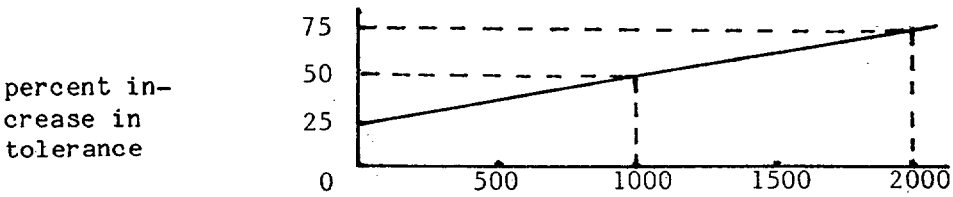
4.4.2.1 Reliability demonstration. Unless otherwise specified, the HARS shall demonstrate a specified MTBF of 750 hours when subjected to the reliability test in accordance with Test Plan III and Test Level E of MIL-STD-781, except that the vibration level shall be limited to one "g" and shall be initiated within 15 minutes of the time the equipment is turned "ON". During the "ON" cycle, the complete HARS minus the compass controller, shall be exercised by a 3-axis, 15° total movement, 5 to 7 cpm reversing scorsby action and by the vibration specified in MIL-STD-781. The compass controller shall also be vibrated as specified in MIL-STD-781. The reliability test cycling period shall not exceed 300 hours per unit. The performance tests shall consist of control set tests (4.5.1.d) minus voltage transients (4.6.4.14), compensation (4.6.4.20) and compass calibration compatibility (4.6.4.24). The DG shall demonstrate an average drift of 1° per hour when averaging the drift in all four cardinal headings with the drift in any one heading not exceeding 2° per hour, and the VG shall demonstrate an average drift of 7° per hour when averaging the drift in all four cardinal headings with the drift in any one heading not exceeding 14° per hour. Drift bias adjustment may be accomplished in addition to the correction action of any relevant failure and the adjustment shall not be considered an additional relevant failure. All failures are considered relevant until proven otherwise to the procuring activity. An out-of-tolerance condition where total failure or parts failure has not occurred, shall be considered a relevant failure if it occurs in two successive performance test periods. If an out-of-tolerance condition occurs in the last performance test period, it will be counted as a

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:54Z
Check the source to verify that this is the currentversion before use.

failure, or the test may be continued for an additional 48 unit hours when all performance tests will be rerun to verify non-recurrence of the out-of-tolerance condition. Accept-reject criteria shall be based on Test Plan III of MIL-STD-781. A thermal survey of the HARS shall be conducted in accordance with MIL-STD-781 to establish points of the highest thermal inertia which shall determine the temperature profile to be used in the test. Results of the thermal survey shall be submitted to the procuring activity for approval prior to starting the demonstration test. In the event of a reject decision, the test shall be immediately stopped, a detailed failure analysis conducted to determine the cause of failures, corrective action changes incorporated in the demonstration units and the test rerun to verify the achievement of the required MTBF. At any time during the reliability test, the contractor may elect to stop the test to incorporate corrective action. If the test is continued, all prior test time and failures shall be counted. Alternately, the test may be restarted at zero time and zero failures at the option of the contractor except that second and subsequent restarts shall require the concurrence of the contracting activity. Pattern failures must be corrected regardless of the outcome of the test. At the conclusion of the test, any corrective action required as a result of the test, shall be incorporated into all the first article test units (all 7 systems) and shall be accomplished at no additional cost to the Air Force.

4.4.2.2 Longevity. At the completion of the reliability test specified in 4.4.2.1, two HARSs shall continue to be tested for a total of 2,500 hours each in the same environment as that required in the reliability test. Components that have been subjected to the reliability test and have individually accumulated 2,500 hours or longer need not be subjected to this test, since the longevity requirement will be considered satisfied if no pattern or wearout failure occurs. During this test, the reliability failure criteria specified herein will not apply; however, the HARS shall not exhibit pattern or wearout failures. Acceptance for delivery of HARSs will be granted after successful completion of all tests specified herein except longevity. (see 6.4).

4.4.2.3 Allowances and failures. The increase in tolerances allowed for reliability testing shall be as follows:



Out-of-tolerance calibration-type conditions shall only be considered as failures if they occur in two successive cycles. However, if an out-of-tolerance condition occurs during the last test period, it shall be considered as a relevant failure or the reliability test continued until the subsequent test period when it can be determined if the out-of-tolerance occurs again or the condition disappears. Except for the above reservations, a failure will be considered as any electrical or mechanical condition which fails to meet the requirements specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

4.4.2.4 Valid time. Component operating time accumulated during tests preliminary to reliability cycling shall not be included in the total required reliability test time.

4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Operational performance tests.

b. Environmental tests.

4.5.1 Operational performance tests. Each HARS or component, as applicable, shall be subjected to the following tests as described under 4.6:

a. Displacement gyro

(1)	Examination of product	(4.6.1.1)
(2)	Burn-in	(4.6.1.2)
(3)	Insulation test	(4.6.1.3)
(4)	Initialization	(4.6.1.4)
(5)	Normal erection rate	(4.5.1.5)
(6)	Fast erection rate	(4.5.1.6)
(7)	Pitch erection cut off	(4.6.1.7)
(8)	Vertical gyro drift rate (spin axis vertical)	(4.6.1.8)
(9)	Roll and pitch drift rate (spin axis deflected)	(4.6.1.9)
(10)	Bias drift adjustment	(4.6.1.10)
(11)	Azimuth drift rate	(4.6.1.11)
(12)	Roll and pitch rotation sensing	(4.6.1.12)
(13)	Roll erection isolation	(4.6.1.13)
(14)	Pitch erection isolation	(4.6.1.14)
(15)	Azimuth rotation sensing	(4.6.1.15)
(16)	Leak rate	(4.6.1.16)
(17)	Weight and dimensions	(4.6.1.17)

b. Electronic control amplifier

(1)	Examination of product	(4.6.2.1)
(2)	Burn-in	(4.6.2.2)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the currentversion before use.

(3) Insulation test	(4.6.2.3)
(4) Roll output	(4.6.2.4)
(5) Pitch output	(4.6.2.5)
(6) Heading output	(4.6.2.6)
(7) Rate of turn output	(4.6.2.7)
(8) Earth rate compensation	(4.6.2.8)
(9) Heading control	(4.6.2.9)
(10) Fast synchronization	(4.6.2.10)
(11) Annunciator control	(4.6.2.11)
(12) Compensation	(4.6.2.12)
(13) Roll erection and heading slaving control	(4.6.2.13)
(14) Roll erection and slaving cutoff inhibit	(4.6.2.14)
(15) Linear acceleration control	(4.6.2.15)
(16) Manual fast erection control	(4.6.2.16)
(17) MAD excitation	(4.6.2.17)
(18) Weight and dimensions	(4.6.2.18)

c. Compass system controller

(1) Examination of product	(4.6.3.1)
(2) Insulation test	(4.6.3.2)
(3) Mode selector switch	(4.6.3.3)
(4) Latitude correction control	(4.6.3.4)
(5) Synchronizer and heading set	(4.6.3.5)
(6) Synchronizing indicator	(4.6.3.6)
(7) MAG VAR control	(4.6.3.7)
(8) Weight, Dimensions and Lighting	(4.6.3.8)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

d. Control assembly. Unless otherwise specified (see 6.2) one control assembly selected at random from each 50 systems on contract shall be subjected to the following tests:

(1)	Initial vertical gyro erection	(4.6.4.1)
(2)	Roll performance	(4.6.4.2)
(3)	Pitch performance	(4.6.4.3)
(4)	Heading performance	(4.6.4.4)
(5)	Fast synchronization	(4.6.4.5)
(6)	Vertical gyro erection rate	(4.6.4.6)
(7)	Vertical gyro manual fast erect	(4.6.4.7)
(8)	Power interruption	(4.6.4.8)
(9)	Earth rate corrected drift	(4.6.4.9)
(10)	Power consumption	(4.6.4.10)
(11)	Pitch erection and slaving cut off	(4.6.4.11)
(12)	Roll erection and heading slaving control	(4.6.4.12)
(13)	Electrical power variation	(4.6.4.13)
(14)	Voltage transients	(4.6.4.14)
(15)	System static accuracies	(4.6.4.15)
(16)	Roll performance	(4.6.4.16)
(17)	Pitch performance	(4.6.4.17)
(18)	Heading performance	(4.6.4.18)
(19)	Low field strength test	(4.6.4.19)
(20)	Compensation	(4.6.4.20)
(21)	Compass controller lighting	(4.6.4.21)
(22)	Rate of turn output	(4.6.4.22)
(23)	Validity signals	(4.6.4.23)
(24)	Compass calibration compatibility (see 6.2)	(4.6.4.24)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

4.5.2 Environmental tests. Unless otherwise specified (see 6.2) three control assemblies shall be selected at random from the first 15 on the contract or order and subjected to the following tests:

a.	Control assembly operational performance tests	(4.5.1.d)
b.	Temperature altitude	(4.6.5.1.)
c.	Humidity	(4.6.5.2)
d.	Fungus	(4.6.5.3)
e.	Rain	(4.6.5.4)
f.	Salt fog	(4.6.5.5)
g.	Dust	(4.6.5.6)
h.	Temperature shock	(4.6.5.7)
i.	Explosive atmosphere	(4.6.5.8)
j.	Vibration	(4.6.5.9)
k.	EMI	(4.6.5.10)
l.	Acoustic noise	(4.6.5.11)
m.	Acceleration	(4.6.5.12)
n.	Shock	(4.6.5.13)
o.	Compatibility	(4.6.5.14)

4.5.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and the cause of failure are determined.

4.5.3.1 Operational performance tests may continue. For operational reasons, operational performance tests may be continued pending investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3.2 Corrective action and retest. Corrective action and retest shall be required for each failure that can be attributed to poor manufacture or design practices or that adversely affects mission completion.

4.5.4 Production reliability test (PRT). Each displacement gyro and electronic control amplifier shall be subjected to a burn-in test specified in 4.6.1.2 and 4.6.2.2. Following burn-in, the displacement gyro and electronic control amplifier with the addition of a compass controller shall be mated to form a complete control assembly and shall be subjected to Test Plan XXIX of MIL-STD-781 for two (2) additional temperature cycles as defined in 4.6.1.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

and 4.6.2.2. These 2 cycles shall be PRT. When a failure occurs during the two PRT cycles, it shall be classified in accordance with MIL-STD-781. The control assembly failed LRU shall then be repaired, and the repair shall be verified by test. The repair verification time shall not be counted toward PRT time. When repair verification has been completed, PRT cycling shall be resumed until two or more PRT cycles have been accumulated failure free. The control assembly shall then be subjected to the control assembly operational performance tests and delivered after successful completion. Failure during the operational performance tests shall be repaired and recorded but will not require additional cycling. All PRT time and relevant failures shall be plotted on a chart. If the reject line is intersected, further delivery of control assemblies shall be discontinued until corrective action is implemented in accordance with MIL-STD-781. The occurrence of a pattern failure shall result in the same actions as does intersection of the reject line. Each control assembly which passes the operational performance tests under the conditions described above shall be considered acceptable. Test records shall be maintained as specified in MIL-STD-781 and the plotted data shall be submitted to the procuring activity on a monthly basis.

4.5.5 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods.

4.6.1 Displacement gyro. The displacement gyro shall be properly connected so that all necessary signals are supplied and all output circuits are correctly loaded. The following tests shall then be performed.

4.6.1.1 Examination of product. The displacement gyro shall be examined for compliance with the requirements specified herein with respect to materials, workmanship, and marking. The case shall be inspected for proper painting and general good workmanship and the connector shall be inspected to make sure that the pins are not bent or damaged.

4.6.1.2 Burn-in. The displacement gyro shall be properly connected so that the outputs and inputs are loaded to simulate actual use. The displacement gyro shall be mounted on a Scorsby-type mechanism which provides a 15° movement at 5 to 7 oscillations per minute. The following conditions shall be applied:

a. Test per MIL-STD-781, Test level E except that vibration exposure is 1 $\pm$0.2g at a nonresonant frequency between 55 and 60 Hz for 10 minutes of every power-on hour.

b. Cycling time, 8 hours per cycle, total 10 cycles, off time of 3 hours, on time of 5 hours.

c. Last 2 cycles of the 10 cycles must be failure free.

4.6.1.3 Insulation test. Insulation resistance of the displacement gyro circuits shall be checked with a megohm meter. A voltage of 500 Vdc shall be applied between the chassis and isolated circuits. The minimum measured resistance shall be 50 megohms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

4.6.1.4 Initialization. One minute after power is applied, the gyro shall be subjected to a Scorsby motion of 3° total excursion. After 2.5 minutes from application of power, roll and pitch erection shall be completed and roll and pitch verticality error shall not exceed ±.25°.

4.6.1.5 Normal erection rate. The displacement gyro shall be mounted on a table, level to within ±.05°, which can be positioned on a N-S axis and an E-W axis. With the gyro positioned N-S and with a 3° total scorsby motion, the pitch gimbal erection rate shall be measured in both + and - directions. The erection rate shall be 1.5 ±.5° per minute. With the gyro mounted in the E-W direction, the test shall be repeated for roll.

4.6.1.6 Fast erection rate. The displacement gyro shall be mounted and subjected to Scorsbied motion as described in 4.6.1.5. The pitch and roll fast erection rates shall be measured in the same manner as the normal rates described in 4.6.1.5. The rate shall be 24° ±4° per minute.

4.6.1.7 Pitch erection cutoff. The displacement gyro shall be mounted on a table level to within 0.05° and positioned to a North heading. The output at pin 3J1-51 with respect to 3J1-50 shall be 0.735 Vac ±0.07V when the gimbal is precessed 4.9° from its erect position. The test shall be repeated in the opposite direction.

4.6.1.8 Vertical gyro drift rate. (Spin axis vertical) The displacement gyro shall be mounted on a table level to within ±25°, headed East. With a 3° total Scorsby motion applied, the roll erection shall be removed and the roll drift recorded for 15 minutes. It shall not exceed 6° per hour. The gyro shall be rotated to a North heading and the gyro allowed to re-erect. The pitch erection shall be removed and the test repeated for the pitch axis.

4.6.1.9 Roll and pitch drift rate - (spin axis deflected). The displacement gyro while headed North shall be mounted on the table level to within ±.25°. With the spin axis tilted 20° east, roll and pitch erection removed, and a total 3° Scorsby motion applied, the roll and pitch drift shall be observed for 5 minutes. The roll and pitch drift rate shall not exceed 10° per hour The test shall be repeated with gyro headings of East, South, and West.

4.6.1.10 Bias drift adjustment. Rotate the DG bias adjustment to the extreme cw position and measure the azimuth displacement of 6 minutes. The bias adjustment shall then be rotated fully ccw and the azimuth displacement measured for 6 minutes. The drift from the full cw to ccw adjustment shall change by 4.5° per hour minimum. Reset the bias adjustment to meet the DG drift requirements of this specification. The bias shall not be reset during the following test (4.6.1.11).

4.6.1.11 Azimuth drift rate. The displacement gyro shall be mounted on a table level to within ±.25° at any random heading. Ten minutes after power has been applied and with a 3° total Scorsby motion, the measurement of azimuth drift shall be started. The drift shall be measured for 1 hour and shall not exceed 1° per hour when earth rate is analytically removed.

4.6.1.12 Roll and pitch rotation sensing. With the platform mounted on a table level to within .05° and erected to within .25° of vertical, the synchro null voltages, line-to-line, shall measure .075 Vrms maximum. With the attitude indicator properly connected and synchro excitation of ϕC, the following phasings shall apply:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Roll - Right wing down shall read increasing angles.

Pitch - Pitch up shall read increasing angles.

4.6.1.13 Roll erection isolation. With the platform pitched up to 60° ±5°, the roll gimbal shall be level to within 0.5°.

4.6.1.14 Pitch erection isolation. With the platform rolled to 60° ±5°, the pitch gimbal shall be level to within 0.5°.

4.6.1.15 Azimuth rotation sensing. With the condition described in 4.6.1.12 the azimuth resolver sin output shall measure 0.075 Vrms maximum at null. A right turn shall read an increasing angle.

4.6.1.16 Leak rate. The displacement gyro shall not lose filling medium at a rate greater than 10^{-4} cm^3/s. The measurement shall be taken at a pressure differential equal to a minimum of 100,000 feet altitude.

4.6.1.17 Weight and dimensions. The displacement gyro shall be examined to determine compliance to figure 2 for dimensions and paragraph 3.3.2.6.9 for weight.

4.6.2 Electronic control amplifier. The electronic control amplifier shall be properly connected so that all necessary signals are supplied and all output circuits are correctly loaded. The following tests shall be performed.

4.6.2.1 Examination of product. The electronic control amplifier shall be examined for compliance with the requirements specified herein with respect to materials, workmanship, and marking. The case shall be inspected for proper painting and general good workmanship and the connector shall be inspected to make sure that the pins are not bent or damaged.

4.6.2.2 Burn-in. The electronic control amplifier shall be properly connected so that the outputs and inputs are loaded to simulate actual use. The following requirements shall apply:

a. Test conditions per MIL-STD-781, test level F.

b. Cycling time, 8 hours per cycle, total of 10 cycles, off time of 3 hours, on time of 5 hours.

c. Last 2 cycles of the 10 cycles must be failure free.

4.6.2.3 Insulation test. Insulation resistance of the electronic control amplifier circuits shall be checked with a megohm meter. A voltage of 500 Vdc shall be applied between the chassis and isolated circuits. The minimum measured resistance shall be 50 megohms.

4.6.2.4 Roll output. There shall be continuity between the following pin combinations:

1J1		1J2
14	to	35
15	to	36
16	to	37

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

4.6.2.5 Pitch output. There shall be continuity between the following pin combinations:

1J1		1J2
11	to	32
12	to	33
13	to	34

4.6.2.6 Heading output. With the amplifier in the DG mode, a continuous range resolver connected to the azimuth input shall be rotated through 360°. The resolver input shall simulate the platform output resolver. The amplifier heading output, as measured on the API, shall follow smoothly. The line-to-line voltage at the null positions shall be 0.075 Vac maximum. The input resolver shall be indexed in 15° steps through 360°. The error between the input and output, when the output is loaded with three control transformers (AY500-5, or equivalent), shall not exceed 21 minutes at any heading.

4.6.2.7 Rate-of-turn output. With the rate-of-turn output loaded with 500 ohms and a continuous resolver connected to the azimuth resolver input, the following outputs shall be measured when the input resolver is rotated in cw and ccw directions:

Rate of Turn (Degrees per Minute	Output mA (dc)
0	0.0 ±0.060
45	0.5 ±0.060
90	1.0 ±0.060
180	2.0 ±0.13
360	4.0 ±0.26

The azimuth resolver simulator shall be rotated at a rate of 3° per minute. The perceptible change in the output signal shall be at least 6 microamps.

4.6.2.8 Earth rate compensation.

4.6.2.8.1 Heading. The signals listed in Table II shall be supplied to the earth's rate input at pin 1J2-49 hi with respect to 1J2-54 lo (+13 Vdc). The output at pin 1J2-26 hi with respect to 1J2-24 lo (+13 Vdc) shall comply with the values listed in Table II.

4.6.2.8.2 Vertical. A latitude signal equivalent to 0° latitude shall be supplied to the VG earth rate corrected (ERC) input at pin 1J2-48. With the heading angle and MAG VAR simulator switch set to the positions described, the roll and pitch outputs measured with respect to 1J1-50 at pins 1J2-22 and 1J2-14 respectively, shall comply with the values listed in TABLE III.

4.6.2.9 Heading control.

4.6.2.9.1 Slaved mode heading control. With the magnetic azimuth detector MAD (at standard field strength) connected to the heading input, the amplifier shall be set in the SLAVED mode and synchronized. The detector shall be rotated 5° in a positive direction. The slaving torquer amplifier output shall measure 5.15 ±1.9 Vdc at 1J2-24 with respect to 1J2-26. The test shall be repeated with the detector rotated 5° in the negative direction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

4.6.2.9.2 DG mode heading control. With the heading output properly loaded and the amplifier in the DG mode, a set heading signal shall be applied. The set heading signal shall be varied to obtain output heading rates of +30° per second, 0, and -30° per second. It shall be possible to obtain a minimum rate of 2° ±0.25° per second. There shall be no effect upon the heading output when a magnetic azimuth detector connected to the amplifier detector input is varied ±180°.

4.6.2.10 Fast synchronization. With the same connections described in 4.6.2.9.1 the detector shall be rotated 180° ±5°. The heading output shall synchronize to within 1° of the detector heading within 6 seconds when fast synchronization is commanded. The gyro azimuth simulator shall be rotated 180° ±5° with the amplifier in the DG mode. When the system is switched to the SLAVED mode, the heading output shall synchronize to within 1° of the detector heading within 6 seconds. The system heading shall then be set 180° ±5° from the detector heading. With the amplifier in the SLAVED mode, the power shall be off long enough for the system to re-initialize. When the power is turned on, the output heading shall synchronize to within 1° of the detector heading within 6 seconds, after completion of the start cycle.

4.6.2.11 Annunciator control. With the same connnections and initial conditions as described in 4.6.2.9.1, the detector shall be rotated 5° in a positive direction. The output at 1J2-62 with respect to 1J2-65 when loaded with 100 ohms, shall be 0.05 ±0.01 Vdc. The test shall be repeated with the detector rotated 5° in a negative direction and negative readings recorded.

4.6.2.12 Compensation. When one-cycle and two-cycle adjustments are being made, the coriolis input signal shall be set to zero.

4.6.2.12.1 One cycle and coriolis compensation. With the MAD detector simulated by three ammeters having 1K resistance maximum, the current flow from each pin to the MAD detector common shall be checked. Before the test, the N-S and E-W adjustment potentiometers shall be adjusted for zero current into the simulated MAD detector.

4.6.2.12.1.1 N-S adjustment. After the N-S adjustment has been rotated fully clockwise and fully counterclockwise, the current magnitudes shall be the equivalent of ±8° minimum compensation. Full scale rotation of the E-W adjustment shall not result in a current change greater than the equivalent of ±1.5°.

4.6.2.12.1.2 E-W adjustment. After the E-W adjustment has been rotated fully clockwise and fully counterclockwise, the current magnitudes shall be the equivalent of ±8° minimum compensation. Full scale rotation of the N-S adjustment shall not result in a current change greater than the equivalent of ±1.5°.

4.6.2.12.1.3 Coriolis signal. With a 90° latitude coriolis input signal applied, the voltage at 1J1-74 shall be 2.45 ±.15 Vdc.

4.6.2.12.2 Two-cycle adjustment range.

4.6.2.12.2.1 E potentiometer. With synchronization commanded, and the MAD detector simulator set at zero degrees, rotation of the E potentiometer from its extreme cw to ccw position shall cause a heading change of ±2° minimum.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:54Z
Check the source to verify that this is the currentversion before use.

Rotation in the cw direction shall cause the heading to increase. Adjustment shall be repeatable to within ±9 minutes.

4.6.2.12.2.2 D potentiometer. With synchronization commanded, and the MAD detector simulator set at 45°, rotation of the D potentiometer from its extreme cw to ccw position shall cause a heading change of ±2° minimum. Adjustment shall be repeatable to within ±9 minutes.

4.6.2.13 Roll erection and heading slaving control. With the amplifier in the slave mode, the latitude signal at 0°, the hemisphere switch at N, the detector simulator shall be positioned to insure slaving. The azimuth continuous resolver shall be rotated in the plus direction at a rate of 3° per minute. Roll erection and slaving shall remain on. The test shall be repeated for the minus direction.

4.6.2.13.1 Switching rates. The test of 4.6.2.13 shall be repeated except the heading resolver simulator shall be rotated at rates of 8°, 30°, and 180° per minute. The roll erection voltage and heading slaving signal shall be cut off within 30, 11, and 8 seconds respectively. Roll erection and heading slaving shall return within 20 seconds after the rotation of the continuous resolver is halted. The test shall be repeated for the minus direction.

4.6.2.13.2 Azimuth oscillation. The test of 4.6.2.13 shall be repeated except with zero-state rate, the heading resolver simulator shall be driven to produce an azimuth oscillation of ±0.8° amplitude with a 1 second period. Roll erection voltage and heading slaving shall remain on. This test shall be repeated with an azimuth oscillation of ±2° amplitude with an 8 second period. Roll erection voltage and heading slaving shall remain on.

4.6.2.13.3 Steady-state rate and azimuth oscillation. The test of 4.6.2.13 shall be repeated except the heading resolver transmitter shall be driven to produce an azimuth oscillation of ±0.8° with a 1 second period superimposed on a steady-state rate of turn of 30° per minute. Roll erection voltage and heading slaving shall be cut off within 30 seconds and shall remain off until the steady-state rate of turn has been removed. The time required to return to normal condition shall not be more than 20 seconds.

4.6.2.14 Roll erection and slaving cutoff inhibit. With the amplifier in the DG mode, the sync switch shall be depressed and rotated to the extreme cw position. With the heading changing at a rate of 30° per second or greater, roll erection shall remain ON, and slaving shall remain OFF.

4.6.2.15 Linear acceleration control. With a 0.700 ±.007 Vac applied to pitch accelerometer input, pitch erection shall remain on. With 0.760 ±0.007 Vac applied to the pitch accelerometer input, pitch erection shall cut off.

4.6.2.16 Manual fast erection control. With .1 Vac ØC applied to the roll and pitch accelerometer inputs (1J2-16 and 1J2-17 with respect to 1J2-15) and manual fast erection activated (1J1-5 connected to 1J1-50), the roll and pitch fixed phase outputs (1J2-46 and 1J2-13 with respect to 1J1-50) shall be 55 ±5 Vac and the roll and pitch control phase outputs (1J2-21 and 1J2-38 with respect to 1J1-50) shall be 18 ±3 Vdc. With roll cut-out activated (1J2-57 connected to 1J1-50) the roll fixed phase output (1J2-46 to 1J1-50) shall be 15 ±2 Vac and the roll control phase output (1J2-21 to 1J1-50) shall be 0 ±.1 Vac. With the roll cut-out not activated (1J2-57 not connected to 1J1-50) and

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

the accelerometer inputs reduced to 0.05 Vac, the roll and pitch control phase outputs (1J2-22 and 1J2-14 to 1J1-50) shall be 7 ±3 Vac.

4.6.2.17 MAD excitation. The signal at pins 1J1-22 (HI) and 1J1-23 (LO), when properly loaded, shall be 23.5 ±2 Vrms, 400 Hz ØC.

4.6.2.18 Weight and dimensions. The electronic control amplifier shall be examined to determine compliance to figure 4 for dimensions and paragraph 3.3.2.7.27 for weight.

4.6.3 Compass system controller. The compass system controller shall be properly connected so that all necessary signals are supplied and all outputs are correctly loaded. The following tests shall then be performed.

4.6.3.1 Examination of product. The compass system controller shall be examined for compliance with the requirements specified herein with respect to materials, workmanship, and markings. The panel, knobs, and sync indicator shall be inspected for proper painting, knob and button actuation, and good workmanship. The connector shall be inspected to make sure that the pins are not bent or damaged.

4.6.3.2 Insulation test. Insulation resistance of the applicable controller circuits shall be checked with a megohm meter. A voltage of 500 Vdc shall be applied between the chassis and isolated circuits. The minimum measured resistance shall be 50 megohms.

4.6.3.3 Mode selector switch. With the mode selector switch in the SLAVED mode of operation, pin 5J1-20 shall be grounded. With the mode selector switch in DG mode, pin 5J1-20 shall be open.

4.6.3.4 Latitude correction control. With the hemisphere selector switch to North and the latitude correction potentiometer to the specified latitude settings, the output shall be as follows:

Latitude (degrees)	Sine Output (volts) (13 Vdc Excitation) Pin 5J1-13 to -7		Cos Output (volts) (5 Vdc Excitation) Pin 5J1-12 to -14	
0.0	0	±0.065 Vdc	5	±0.025 Vdc
30.0	6.5	±0.065 Vdc	4.33	±0.025 Vdc
60.0	11.26	±0.065 Vdc	2.5	±0.025 Vdc
90.0	13.0	±0.065 Vdc	0	±0.025 Vdc

4.6.3.5 Synchronizer and heading set. With the system in the DG mode, the sync switch shall be depressed and rotated from one extreme to the other. The set heading output voltage shall be continuously variable from 7.1 ±.8 Vac (ØC + 90°) to 7.1 ±.8 Vac (ØC - 90°). With the mode selector switch in the SLAVED mode, depress the sync switch. Pin 5J1-22 shall be grounded.

4.6.3.6 Synchronizing indicator. A heading error signal of 0.025 mA applied to the synchronizing indicator shall show a perceptible deflection of the needle. A heading error signal of 0.5 mA applied to the synchronizing indicator shall show a full scale deflection of the needle.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

4.6.3.7 MAG VAR control. With the switch in the +15° position, there shall be continuity between pins 16 and 17 of 5J1. With the switch in the -15° position, there shall be continuity between pins 3 and 17 of 5J1. With the switch in the 0° position, there shall be no continuity between pins 16 and 17 and between pins 3 and 17 of 5J1.

4.6.3.8 Weight, dimensions and lighting. The compass controller shall be examined to determine compliance to figure 6 for dimensions and paragraph 3.3.2.8.11 for weight. The controller shall be placed side by side with the prime standard and shall meet the lighting requirements of 3.3.2.8.1 with 5.0 ±0.1 Vdc and 2.85 ±0.05 Vdc applied to the lighting circuits of each unit.

4.6.4 Control assembly. The control assembly shall be properly connected so that all necessary signals and power are supplied and all outputs are properly loaded. The magnetic azimuth detector (MAD) simulator shall be properly connected to the electronic control amplifier. The compass controller shall be set to the local latitude and in the SLAVED mode of operation. The MAD simulator and the displacement gyro case heading shall be set to 0°. The following tests shall be performed.

4.6.4.1 Initial vertical gyro erection. The control assembly shall be set up as in 4.6.4 and power and the appropriate signal loading shall be applied. The vertical gyro shall be erected as specified in 4.6.1.4.

4.6.4.2 Roll performance. The displacement gyro shall be mounted on a tilt table and power shall be applied. The gyro shall be displaced in the roll axis from -90° to +90° with the tilt table. The roll output, as observed on the API, shall follow in the correct direction smoothly and continuously. No error shall exceed 1° during the displacement.

4.6.4.3 Pitch performance. The displacement gyro shall be mounted on a tilt table so that the pitch axis is parallel to the tilt axis and the gyro displaced in the pitch axis from -60° to +60°. The pitch output shall follow smoothly and continuously in the correct direction with no hunting and jumping. The errors during the displacement shall not exceed 1°.

4.6.4.4 Heading performance. The displacement gyro shall be mounted on a turntable in the normal operating position and the turntable rotated in a cw direction for 360°. The transmitted heading signal from the amplifier shall increase by 360°. The turntable shall be rotated in a ccw direction for 360°; the transmitted heading signal from the amplifier shall decrease by 360°. In both cw and ccw rotation, the heading error shall not exceed 1°.

4.6.4.4.1 Heading coupler in SLAVED mode. With the mode selector switch in the SLAVED mode, the heading on the detector simulator shall be changed 5° ahead of the output heading. The sync indicator shall indicate 3/4 to full scale deflection in the right direction. The heading shall increase at 1.5° ±0.5° per minute. The sync indicator shall indicate an out-of-sync condition in the right direction until the output of the gyro is synchronized to the output of the detector simulator at which time, the sync indicator needle shall be centered to within ±0.0625 inch. The above test shall then be repeated in the negative direction.

4.6.4.4.2 DG mode. With the mode selector switch in the DG mode, the magnetic detector shall be rotated ±180°. There shall be no movement of the heading output except for normal gyro effects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

4.6.4.4.3 Heading set control. With the system in the DG mode, the heading set knob shall be depressed and rotated in the cw direction. The heading output shall increase smoothly at a rate of 30° per second or greater when the knob is at its extreme position. The test shall be repeated for the ccw direction, and the heading shall decrease at the above rate.

4.6.4.5 Fast synchronization. With the system in the SLAVED mode and the heading on the detector simulator set 180° ±2° from the output heading, the push-to-sync control shall be actuated. The system shall be synchronized to within 1° of the detector setting within 6 seconds. The sync indicator shall be centered within ±0.0625 inch after the output of the gyro is completely synchronized with the output of the detector.

4.6.4.6 Vertical gyro erection rate. With the displacement gyro headed North, the pitch erection shall be removed and the pitch gimbal precessed 3° in a positive angle direction. The erection shall be reapplied and the gimbal shall erect at 1.5 ±0.5° per minute. The test shall be repeated from a negative angle direction. The gyro shall then be rotated to a West heading and the test repeated for roll.

4.6.4.7 Vertical gyro manual fast erect. With the displacement gyro in a North heading, the pitch erection shall be removed and the pitch gimbal precessed 20° in a positive angle direction. A manual fast erection signal shall be supplied and the gimbal shall erect at 24° ±4° per minute. Within 0.5 ±0.25° of null, erection shall drop to the normal rate. The test shall be repeated from the negative direction. The gyro shall then be rotated to a West heading and the above test repeated for roll.

4.6.4.8 Power interruption.

4.6.4.8.1 Vertical output. The vertical output shall be checked by the following procedures. Power shall be removed with the displacement gyro erected on a table level to within 0.5°. The gyro shall be displaced through +15° roll, -15° roll, +15° pitch, -15° pitch, and then level. Power shall be reapplied not more than 30 seconds and not less than 25 seconds after power has been turned off. The pitch and bank axes shall indicate level to ±1° within 10 seconds after reapplication of power.

4.6.4.8.2 Heading output. With the displacement gyro erected on a table level to within 0.5°, power shall be removed for a period of 10 seconds. During this period, the gyro shall be rotated to +5°, then -5°, and then to 0.25° of the original heading. The heading after the power interruption is removed shall not differ by more than 1° from the heading output prior to power interruption.

4.6.4.8.3 Displacement gyro. With the displacement gyro on a tilt table level to within 0.5° and erected, the displacement gyro shall be rotated 45° in pitch and roll. Power shall be removed long enough to cause recycling when power is reapplied. Power shall be applied within 25 seconds of the time required for cycling. The displacement gyro shall initialize properly in accordance with 4.6.4.1 with no damage.

4.6.4.8.4 Power shutdown. Power shall be removed with the displacement gyro on a rate table and erected. Five minutes after power removal, the gyro shall be rotated at a rate of 30° per second for at least 180° in pitch, roll and

Source: https://assist.dla.mil -- Downloaded: 20260-26T04:54Z
Check the source to verify that this is the current version before use.

azimuth. No damage shall occur. Power shall be reapplied and the gyro re-erected per 4.6.4.1. The test shall be repeated after a 10-minute power shutdown, and again after a 15-minute power shutdown.

4.6.4.9 Earth rate corrected drift.

4.6.4.9.1 Heading. With the displacement gyro on a Scorsby table with a 3° total displacement, the latitude correction dial set to the local latitude, and the mode switch on the DG mode, a drift test shall be run on two cardinal headings. The test shall be run for at least 30 minutes on each heading and the drift rate shall be equal to or less than 1.0° per hour.

4.6.4.9.2 Vertical.

4.6.4.9.2.1 MAG VAR 0°. With the displacement gyro on a Scorsby table with a 3° total displacement, the latitude control shall be set to the local latitude, the mode selector switch to the SLAVED mode, and the MAG VAR switch to 0°. The gyro shall be placed to a North heading and the magnetic detector simulator to 0°. The gyro shall be allowed to drift for 15 minutes in each axis. The drift shall not exceed 7° per hour. The gyro shall be re-erected, the magnetic detector simulator and platform rotated to 90°, and the drift test repeated.

4.6.4.9.2.2 MAG VAR +15°. The drift test shall be repeated with the MAG VAR switch set to +15°, and the magnetic detector simulator and gyro to 195°. The magnetic detector simulator and gyro shall then be set to 285° and the drift test repeated.

4.6.4.10 Power consumption. With the system properly connected with the appropriate signal loadings, power shall be applied. Power consumption shall not exceed the following:

a. Starting: 120 VA, 0.7 power factor; 5.5 VA at 5 V for panel lighting.

b. Running (after 2.5 minutes): 75 VA, 0.6 power factor; 5.5 VA at 5 V for panel lighting.

4.6.4.11 Pitch erection and slaving cutoff. With the complete system appropriately connected and in the SLAVED mode, the displacement gyro shall be mounted on a table level to within 0.05° and in a North heading. Power shall be applied and the system allowed to stabilize. The vertical gyro shall be precessed to a pitch-up angle of 4.15°. Pitch erection and slaving shall remain ON. The vertical gyro shall then be precessed to a pitch angle greater than 5.55°. Pitch erection and slaving shall be cut off. The test shall be repeated for the pitch down condition.

4.6.4.12 Roll erection and heading slaving control. With the complete system properly connected, the displacement gyro shall be mounted on a rate table and power applied, allowing the system to stabilize with the system in SLAVED mode. The rate table shall be rotated cw at 3° per minute. Roll erection, pitch erection, and heading slaving shall remain ON. The rate table shall be rotated at 8° per minute. Roll erection and heading slaving shall be cut off within 30 seconds. Pitch erection shall remain ON. The rotation shall be stopped immediately. Roll erection and heading slaving shall be restored within 20 seconds. The 8° rate shall be repeated with the system in manual fast erect. Roll erection shall cut off. The test shall be repeated with the gyro rotated in the ccw direction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

4.6.4.13 Electrical power variation. With the system properly connected, proper loading and nominal power shall be applied and sufficient time allowed for the system to stabilize. The frequency and voltage shall be changed to 380 Hz and 130 V, respectively. While at this frequency and voltage, the system shall be subjected to the tests specified in 4.6.4.2 through 4.6.4.7 and 4.6.4.9. Tests required by 4.6.4.16.2, 4.6.4.17.2, and 4.6.4.18.2, except for voltage measurement, shall be performed. The system shall meet these tests except the limits shall increase 15 percent. The test shall be repeated with frequency and voltage of 420 Hz and 123 V, respectively.

4.6.4.14 Voltage transients. With the system properly connected, proper loading and nominal power shall be applied and sufficient time allowed for the system to stabilize. The voltages and durations specified in 3.3.2.5 shall be applied. A minimum of 200 spikes of 600 V, 50 microsecond duration, shall be applied to the system. The system shall then be subjected to and meet the tests specified in 4.6.4.1 through 4.6.4.4.3.

4.6.4.15 System static accuracies. With the displacement gyro mounted on a Scorsby table with a 3° total movement, the system shall be set in the SLAVED mode, the detector simulator set to 0° heading and the Scorsby table leveled. Without applying power to the scorsby table and within 2.5 minutes after application of power to the system, the pitch and roll outputs shall be within ±0.50° of level, and the heading output shall be 0 ±0.75°. Repeat the initialization cycle with the Scorsby table operational. Within 2.5 minutes after application of power to the system, the pitch and roll outputs shall be within ±0.25° of level, and the heading output shall be 0 ±0.5°, with the table level.

4.6.4.16 Roll performance.

4.6.4.16.1 High-rate followup. The displacement gyro shall be mounted on a rate table so that the roll axis is colinear with the rate table axis, which shall be horizontal. Power shall be applied to the system and the system allowed to stabilize. The pitch and roll outputs shall be 0° to within ±0.5°. The rate table shall be rotated at least two revolutions at 600° per second in the cw direction. The displacement gyro shall be releveled immediately after rotation has stopped. The pitch and roll output shall be 0° within ±0.5° of the initial value. The test shall be repeated for the ccw rotation. When the rate table is rotated at a rate of 2 rpm in the cw direction, the roll output shall follow the rate table position to within ±1° moving in increments of less than 0.1°. The test shall be repeated for the ccw rotation.

4.6.4.16.2 Roll output drive capability. With the roll output loaded with three AY500-5 synchro control transformers or equivalent, the gyro shall be indexed in eight 45° steps from 0° through 315°. The allowable error shall be ±0.35°. The output voltage at the maximum point shall be 11.8 ±0.7 Vrms line-to-line with a phase of 9° ±5° with respect to the excitation phase.

4.6.4.17 Pitch performance.

4.6.4.17.1 High-rate followup. The displacement gyro shall be mounted on a rate table so that the pitch axis is colinear with the rate table axis, which shall be horizontal. Power shall be applied to the system and the system allowed to stabilize. The pitch and roll outputs shall be 0° ±0.5°. The rate

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the currentversion before use.

table shall be rotated at 36° per second for at least one complete revolution. The heading and roll output shall be reversed in phase as the displacement gyro passes through the pitch zenith at 90° and again at 270°. After the loop performance, the displacement gyro shall be immediately releveled and the pitch and roll output shall be within ±0.5° of the initial value. When the rate table is rotated at a rate of 2 rpm, the pitch output shall follow the rate table position to within ±1°, moving in increments of less than 0.1°. The phase reversals of this paragraph shall occur.

4.6.4.17.2 Pitch output drive capability. With the pitch output loaded with three AY500-5 synchro control transformers or equivalent, the gyro shall be indexed in 30° steps between +60° and -60°. The allowable error shall be 0.35°. The output voltage at the maximum point shall be 11.8 ±0.7 Vrms line-to-line with a phase of 9° ±5° with respect to the excitation phase.

4.6.4.18 Heading performance.

4.6.4.18.1 Followup. The displacement gyro shall be mounted on a rate table so that the heading axis is colinear with the rate table axis. With the system in DG mode, power shall be applied and the system allowed to stabilize. The gyro shall be rotated 360° at 36° per second. The table shall be stopped and aligned to its initial heading. The heading output shall be within ±0.75° of its initial value. The table shall be rotated at 2 rpm in a cw direction, and the heading output shall follow the rate table within ±0.5° moving in increments no larger than 0.1°. The test shall be repeated in the ccw direction.

4.6.4.18.2 Output drive capability. With the heading output loaded with three AY500-5 synchro control transformers or equivalent, the gyro shall be indexed in twenty-four 15° steps from 0° through 345°. The allowable error shall be 0.5°. The output voltage at the maximum point shall be 11.8 ±0.7 Vrms line-to-line with a phase of 9° ±5° with respect to the excitation phase.

4.6.4.19 Low-field strength tests.

4.6.4.19.1 Heading coupler action in SLAVED mode. The compass controller shall be set in SLAVED mode and with the system operating properly, the detector simulator simulating a weak field shall be set 5° ahead of the system heading output. The output heading shall slave toward the detector heading at a rate of 1.5° ±0.5° per minute. This slaving rate shall be maintained until the system is synchronized to the detector within approximately ±1.5°. Slaving shall continue at a decreasing rate until the system is synchronized within 0.1°.

4.6.4.19.2 Fast synchronization. The detector simulator shall be set 180° ±5° from the system heading output still simulating a weak field. The set heading button shall be depressed, activating the fast sync and the system shall synchronize to within 1° of the detector within 6 seconds.

4.6.4.20 Compensation. The compensation test shall be performed as follows:

a. Perform one cycle compensation as per 4.6.2.12.1.1 and 4.6.2.12.1.2 except that an actual MAD is used in lieu of a MAD simulator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

b. Coriolis compensation shall be performed as follows: With 90° latitude coriolis input signal applied, the voltage at 1J1-75 to 1J1-78 shall be .5 ±.2 mVdc and the voltage at 1J1-77 to 1J1-78 shall be 1.2 ±.3 mVdc. Three 100 ohm ±1% resistors may be used in lieu of the MAD.

c. Perform 2 cycle adjustment range as per 4.6.2.12.2.1 and 4.6.2.12.2.2.

4.6.4.21 Compass controller lighting. The compass controller shall be tested for brightness, color, contrast, stray light, and dial visibility, and shall meet the requirements specified in 3.3.2.8 and subparagraphs.

4.6.4.21.1 Panel touch temperature. With the compass controller lighted for a period of at least 15 minutes in an ambient temperature of 27°C, the surface temperature for the controls, displays, and knobs shall not exceed 41°C. The surface panel temperature shall be taken at a minimum of six points placed throughout the panel.

4.6.4.22 Rate-of-turn output. The displacement gyro shall be leveled in a normal operating position on a rate-of-turn table and power applied. With the rate-of-turn output loaded with 500 ohms at 0° per minute rate of turn, the rate-of-turn output shall be 0 ±60 uA. With the table rotated at 45° per minute and 180° per minute, the rate-of-turn output shall be 0.5 ±0.06 mA and 2.0 ±0.13 mA, respectively. The turn rate shall be increased until saturation is reached. It shall be greater than 380° per minute but less than 720° per minute.

4.6.4.22.1 Transient response. With the system in the DG mode, the gyro shall be rotated at a rate of approximately 360° per minute in either direction to obtain an output of 4.0 mA. The time required for the output current to drop from 4.0 to 1.47 mA, when the turn rate is suddenly reduced to 0, shall be 1.75 ±0.25 seconds.

4.6.4.23 Validity signals. Continuity between pins 25 and 26 and between pins 27 and 28 of 1J1 shall be broken when the system indicates a non-valid condition.

4.6.4.23.1 Initialization. With the system mounted on a table level to within ±0.05° and in the DG mode, power shall be applied. The application of power shall be considered as time equals 0. At t=25 seconds, the roll or pitch axis shall be maintained at approximately 1.5° by operating the precession erection controls. At t=100 seconds, the system shall be allowed to erect. A non-valid condition shall be indicated until roll and pitch are within 0.5° ±0.025° of the settling position.

4.6.4.23.2 Fast erection, synchronization, and heading slew. The amplifier shall cause a non-valid condition to be indicated immediately when manual fast erect, synchronization, or heading slew controls are operated.

4.6.4.23.3 Power supply. The amplifier shall cause a non-valid condition to be indicated at 3.5 ±1.5 seconds after one of the 3 phase power leads is opened.

4.6.4.23.4 Heading rate generator. The amplifier shall cause a non-valid condition to be indicated at 3.5 ±1.5 seconds after the excitation to the resolver is removed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

4.6.4.23.5 Servo loop. With the platform leveled within ±0.25° and stabilized, pin 59 of 1J2 shall be tied to pin 50 of 1J1. The platform shall be rotated about the roll axis until the signal at pin 46 of 1J1 drops to less than 0.5 Vdc (2° to 4°). After 3.5 ±1.5 seconds, the system shall indicate a non-valid. Returning the system to level shall remove the non-valid indications. After returning to level, the platform shall be rotated in pitch until the above signal again reads less than 0.5Vdc (2° to 4°). After 3.5 ±1.5 seconds, the system shall indicate a non-valid condition. The platform shall be returned to level, removing the non-valid indication, and the short from 1J2-59 to 1J1-50 shall be removed.

4.6.4.23.6 Heading servo. The system shall be placed in the DG mode and 12 VRMS (ØC + 90°) applied to pin 1J2-62. After 3.5 ±1.5 seconds, the system shall indicate a non-valid condition.

4.6.4.24 Compass calibration compatibility. The HARS shall be properly interconnected with a DT-309/AJN Magnetic Azimuth Detector (MAD) and an MC-1M Magnetic Compass Calibrator (MCC) Set. The HARS shall be in the SLAVED mode of operation with the PUSH-TO-SYNC knob depressed; and the latitude shall be set at 0°. The compatibility test shall be performed as follows:

a. N-S deviation range. Using the MCC, apply a 0° heading. Rotate the compensator module N-S adjustment fully cw and ccw while observing the HARS heading output. The heading output shall change 8° minimum for the cw and ccw positions. Return the N-S adjustment to neutral position.

b. E-W deviation range. Using the MCC apply a 90° heading, rotate the compensator module E-W adjustment fully cw and ccw while observing the HARS heading output. The heading output shall change 8° minimum for the cw and ccw positions. Return the E-W adjustment to neutral position.

c. E setting range. Using the MCC apply a 0° heading. Rotate the compensator module E potentiometer from its extreme cw to ccw position while observing the HARS heading output. The cw and ccw rotation shall result in a ±2° minimum heading change. Reset the E potentiometer to a neutral position.

d. D setting range. Using the MCC apply a 45° heading. Rotate the compensation module D potentiometer from its extreme cw to ccw position while observing the HARS heading output. The cw and ccw rotation shall result in a ±2° minimum heading change. Reset the D potentiometer to a neutral position.

e. Coriolis range. Using the MCC apply a 0° heading. The HARS heading output shall be recorded. The latitude control shall then be set to 90°. The HARS heading output shall be recorded. The difference between the HARS heading output for 0° latitude and 90° latitude shall be 0.53° ±0.17°.

4.6.5 Environmental tests. The HARS shall be tested using the environmental test methods designated in the following subparagraphs to verify HARS compliance with the requirements of 3.3.5 and its subparagraphs. The tests may be performed individually or in logical combinations with operational performance tests.

4.6.5.1 Temperature altitude. The HARS shall be subjected to a temperature-altitude test for Class II instruments in accordance with MIL-STD-810, Method 504, Procedure I. Maximum test temperature with the equipment operating shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

be 71°C. Maximum non-operating temperature for the controller shall be 85°C. Testing to determine satisfactory operation shall consist of the test in 4.6.4.1 and observing that the pitch, roll, and heading outputs follow the Scorsby motion in the three axes smoothly without sticking, jumping, or hunting. Tests to determine performance characteristics shall consist of 4.6.4.1, 4.6.4.4.1, 4.6.4.4.2, and 4.6.4.9. The tests of 4.6.4.1 must be completed within 5 minutes. During initialization, a verticality error of 0.75° shall be allowed. In 4.6.4.9, DG drift of 1.5° per hour shall apply. The DG and VG drift test may be performed at one heading only with a MAG VAR setting of 0°. The VG drift shall not exceed 12° per hour, with earth rate compensation applied. For the other tests, a 10 percent increase in tolerance shall apply. At the conclusion of this test, the units shall pass applicable performance tests with no increase in limits.

4.6.5.2 Humidity. The components shall be subjected to a humidity test in accordance with Method 507, Procedure I, of MIL-STD-810. Upon completion of this test, the electronic control amplifier and compass controller shall meet the individual tests described in 4.6.2.6 through 4.6.2.17, and 4.6.3.3 through 4.6.3.6. There shall be no evidence of detrimental deterioration, corrosion, or rusting of external parts of the components. A dummy may be used in place of the hermetically sealed portion of the displacement gyro.

4.6.5.3 Fungus. The components shall be subjected to a fungus test in accordance with Method 508, Procedure I, of MIL-STD-810. Upon completion of this test, there shall be no evidence of detrimental deterioration, corrosion, or fungus growth of any of the components. A dummy may be used in place of the hermetically sealed portion of the displacement gyro.

4.6.5.4 Rain. The electronic control amplifier and compass controller shall be subjected to a rain test in accordance with Method 506, Procedure I, of MIL-STD-810. Upon completion of this test, the components shall meet the individual tests described in 4.6.2.6 through 4.6.2.17 and 4.6.3.3 through 4.6.3.6. There shall be no evidence of water damage.

4.6.5.5 Salt fog. The components shall be subjected to the salt fog test in accordance with Method 509, Procedure I, of MIL-STD-810. Upon completion of the test, the components shall meet the tests specified in 4.6.2.6 through 4.6.2.17 and 4.6.3.3 through 4.6.3.6. There shall be no evidence of physical deterioration. A dummy may be used in place of the hermetically sealed portion of the displacement gyro.

4.6.5.6 Dust. The amplifier and controller shall be subjected to the dust test in accordance with Method 510, Procedure I, of MIL-STD-810. Failure criteria for the dust test shall be failure to meet the performance tests specified in 4.6.2.6 through 4.6.2.17 and 4.6.3.3 through 4.6.3.6.

4.6.5.7 Temperature shock. The components shall be subjected to a temperature shock test in accordance with Method 503, Procedure I, of MIL-STD-810. Upon completion of this test, the components shall meet the tests specified in 4.6.4.1 through 4.6.4.4, 4.6.4.11, 4.6.4.12 and 4.6.4.22.

4.6.5.8 Explosive atmosphere. The electronic control amplifier and the compass controller shall be subjected to an explosive atmosphere test in accordance with MIL-STD-810, Method 511, Procedure I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

4.6.5.9 Vibration. The HARS shall be tested in accordance with MIL-STD-810, Methods 514.1 (normal) and 519.1 (gunfire), compass controller only, except as modified below. The gyro and electronic control amplifier will be mounted on their mounting bases (Figures 3 and 5) and the complete units attached to the test fixture.

4.6.5.9.1 Sinusoidal vibration. A resonance search shall be performed in each axis while cycling from 5-2000-5 Hz at the levels shown in 3.3.5, subparagraph h, for sinusoidal vibration. The sweep time per axis shall be 20 minutes. All major resonant frequencies shall be recorded.

4.6.5.9.2 Random vibration. The equipment shall be subjected to the random vibration requirements of 3.3.5, subparagraph h, along each of the mutually perpendicular axes. For the non-gunfire (normal) spectra, the time shall be one (1) hour per axis for each of the design (functional) and the endurance (fatigue) tests. For each axis, one-half of the design test shall be conducted first, followed by the endurance test, and then the second half of the design test. The random gunfire levels shall be applied for 15 minutes in each axis. The equipment shall meet the performance requirements specified herein during the design testing for both the normal and gunfire spectra.

4.6.5.10 Electromagnetic interference (EMI) tests. The HARS shall be subjected, on a system or on an individual unit basis, to the following EMI tests in accordance with MIL-STD-462:

a. CE03 "Conducted Emission" 20 kHz to 50mHz.

b. CS01 "Conducted Susceptibility" 30 Hz to 50 kHz.

c. CS02 "Conducted Susceptibility" 50 kHz to 400 mHz.

d. CS06 "Conducted Susceptibility" Spike.

e. RE02 "Radiated Emissions" 14 kHz to 10 gHz (The HARS shall be tested for EMI over the frequency range of 150 kHz to 1000 mHz for test method RE02).

f. RS02 "Radiated Susceptibility" magnetic induction field.

g. RS03 "Radiated Susceptibility" 14 kHz to 10 gHz (The HARS shall be tested for susceptibility over the frequency range of 14 kHz to 1000 mHz for test method RS03).

4.6.5.11 Acoustic noise. The control assemblies shall be subjected to an acoustic noise test in accordance with Method 515, Intensity Category B, Procedure I, of MIL-STD-810. The control assembly outputs of pitch, roll, and heading shall not vary more than 0.75° from the initial position established with 10 percent of this noise level and shall return to the initial position within ±0.5° when the noise level is again reduced to 10 percent of the test noise level.

4.6.5.12 Acceleration. The components shall be subjected to acceleration tests in accordance with Procedures I and II of Method 513, MIL-STD-810. At the completion of these tests, the components shall meet the requirements of 4.6.4.1 through 4.6.4.12, except for the tests of 4.6.4.8 and its subparagraphs (power interruption) and 4.6.4.10 (power consumption).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

4.6.5.13 Shock. The components shall be subjected to shock tests to determine their structural integrity and functional adequacy under conditions of service shock. The structural integrity of the equipment, mounting base (Figures 3 and 5), and other holding devices shall be determined by the crash safety test. Each shock impulse shall be of the time duration, shape, and tolerance specified in Figure 516.1-2 of MIL-STD-810, except that the maximum value of acceleration shall be within ±10 percent of the specified value when measured with a 250-Hz low-pass filter.

4.6.5.13.1 Equipment operation (service shock with equipment normally mounted). The equipment shall be attached to the test fixture by the mounting means used in the service installation plus all electrical cables and any other connector necessary for proper operation. The equipment shall be operating satisfactorily during this test. The equipment shall be subjected to 18 impacts. After the impacts on each axis, the equipment shall be thoroughly checked for any failure and a performance check shall be made. The shocks shall be applied in the following directions:

a. First orthogonal axis - three shocks in each direction.

b. Second orthogonal axis - three shocks in each direction.

c. Third orthogonal axis - three shocks in each direction.

4.6.5.13.2 Equipment crash safety (equipment normally mounted). Each component or dummy load representing a gyro, amplifier, and compass controller shall be attached to a test fixture by the mounting means intended for use in the service installation. If a dummy load is used, it shall simulate equipment weight, center-of-gravity location, approximate radii of gyration (internal shock mounts of displacement gyro must be simulated), and attach-point structure. Testing shall be in accordance with the test levels listed in Table IV and the Figure 516.1-2 of MIL-STD-810. The mounting base shall be examined for structural failure. Bending and distortion will be permitted but there shall be no structural failure of the attach points, and the equipment or dummy load shall remain in place. The peak amplitude specified on Figure 516.1-2 of MIL-STD-810 must be ±10 percent of the specified value measured with a 250-Hz low-pass filter. Duration shall be within ±9 percent (11 ±1 milliseconds or 22 ±2 milliseconds) of the specified value.

4.6.5.14 Compatibility. The HARS shall be subjected to a compatibility test, to demonstrate compliance with the requirements of paragraph 3.2.3. When interfaced with the systems described in paragraph 3.2.3, subparagraph a through d, the HARS shall be subjected to and meet tests in paragraphs 4.6.4 through 4.6.4.7. Satisfactory completion of the test specified in 4.6.4.24 shall constitute compliance with subparagraph e of 3.2.3.

4.7 Packaging inspection.

4.7.1 Quality conformance. The inspection of the preservation-packaging and interior package marking shall be in accordance with Group A and B quality conformance inspection requirements, Section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

4.7.2 First article. When the unit container is capable of serving as the shipping container, first article inspection and rough handling tests, as outlined in Section 4 of MIL-P-116, shall be accomplished followed by a functional test of the unit to insure freedom from operational malfunction.

5. PACKAGING.

5.1 Preservation-packaging. Preservation and packaging shall be level A or C as specified by the procuring activity.

5.1.1 Level A.

5.1.1.1 Cleaning. Unit shall be cleaned in accordance with Process C-1 of MIL-P-116.

5.1.1.2 Drying. Immediately after cleaning, the unit shall be dried following any one of combination of the drying procedures of MIL-P-116. The drying procedures employed shall not be injurious to the unit.

5.1.1.3 Preservation-application. Not required.

5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each unit shall be packaged in quantity unit packs of one each in accordance with Method 1A14 of MIL-P-116. Overbox each completed pack in a PPP-B-636 fiberboard, weather-resistant class box. Cushioning material between box and container shall be of a type, density and thickness that will protect the unit from impact shock damage when tested as specified in 4.6.5.13. Insofar as practical, exterior containers shall be of uniform shape, minimum tare and cube consistent with protection required.

5.1.2 Level C. Units shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2.f).

5.2.1 Level A. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-601. Table II of MIL-STD-794 applies for Army use.

5.2.2 Level B. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636, class domestic. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, the shipping containers shall conform to specification PPP-B-636 class weather-resistant.

5.2.3 Level C. Packaged units shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to applicable carrier rules and regulations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

5.3 Marking of shipment. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The shipment marking nomenclature shall be as follows, as applicable:

CONTROL SET, GYROSCOPE, ATTITUDE HEADING AN/ASN-129
GYROSCOPE, DISPLACEMENT CN-1466/ASN-129
BASE MOUNTING NSN 6615-01-013-9525
AMPLIFIER, ELECTRONIC CONTROL, AM-6851/ASN-129
BASE MOUNTING NSN 6615-01-027-9731
CONTROLLER, COMPASS SYSTEM C-10005/ASN - 129

5.3.1 Additional marking. Components prepared for shipment in accordance with Level C/C and intended for use at destination shall be conspicuously marked on two opposite sides of the exterior container:

FOR IMMEDIATE USE

DO NOT STORE IN THIS PACK

6. NOTES

6.1 Intended use. The HARS described herein is intended for use in aircraft to provide roll, pitch, magnetic heading, rate of turn and validity signals, for indication and interface with other aircraft systems. Provisions are made for various types of operation, error compensation and detector calibration depending upon the application and associated equipment.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable contract number.

c. Applicable part number.

d. Items of data required.

e. Location and conditions for first article testing, instructions for submission of preproduction test procedures, test plan, test report and disposition instructions for test samples (see 3.1 and 4.4).

f. Level of packaging and packing required.

g. When control assembly tests (4.5.1.d) will not be conducted.

h. When the environmental tests (4.5.2) will not be conducted.

i. When production reliability test (PRT) of 4.5.4 will not be conducted.

j. When the compass calibration compatibility (4.6.4.24) will not be conducted.

k. When the compatibility testing (4.6.5.14) will not be conducted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

6.3 Abbreviations.

ac	Alternating Current
cpm	Cycles Per Minute
CSC	Compass System Controller
ccw	Counter Clockwise
cw	Clockwise
dB	Decibel
dc	Direct Current
DG	Directional Gyroscope
DA	Double Amplitude
ECA	Electronic Control Amplifier
ETI	Elapsed Time Indicator
Gyro	Gyroscope
HARS	Heading Attitude Reference System (AN/ASN-129)
HUD	Head Up Display
Hz	Hertz (cycles per second)
Hdg	Heading
Lat	Latitude
LRU	Line Replaceable Unit
MAG VAR	Magnetic Variation
MAD	Magnetic Azimuth Detector
MCC	Magnetic Compass Calibrator
MTBF	Mean Time Between Failure
MTTR	Mean Time To Repair
MMH/FH	Maintenance Manhour per Flight Hour
mA	Milliamperes
N/A	Not Applicable
PPL	Preferred Parts List
PRT	Production Reliability Test
lbf/in^2	Pounds Per Square Inch
rms	Root Mean Square
rpm	Revolutions Per Minute
SRU	Shop Replaceable Unit
SE	Support Equipment
Sync Ind	Synchronizing Indicator
TBD	To be determined
TTM	Time Totalizing Meter
Ø	Phase
uA	Microamperes
V	Volt
VA	Volt Ampere
Vac	Volts, Alternating Current
Vdc	Volts, Direct Current
VG	Vertical Gyroscope

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

6.4 Longevity. If wearout or pattern failures occur during the longevity test (see 4.4.2.2) the contractor will be required to correct the design deficiency at no cost to the Government.

TABLE I. Warning interlock functions.

Function	When monitored		Conditions
	Initial start cycle	Continuously	3.5 ±1.5 seconds time delay except where noted
VG spin motor	X		Roll and pitch greater than 0.5° ± 0.25° and 82 seconds have not elapsed
Pitch and roll torquers	X		
Acceleration and erection amplifiers	X		
Slip rings	X		
Power supplies	X		
VG erection circuits	X		
Initialization circuits	X		
Fast erection		X	At signal application
Synchronization		X	
Heading slew		X	No time delay
Primary Power		X	Loss of any phase of 3 phase primary power
Servo loops		X	Error of 2.7° nominal between the sensitive and servo axes
Heading rate generator		X	Loss of platform resolver excitation
Heading valid		X	Error 2.7° nominal between azimuth input and servo heading output

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE II. Heading earth's rate compensation.

Earth Rate Input	Earth Rate Output
0 Vdc	0 ±0.01 Vdc
+13 Vdc	-0.818 Vdc ±0.025 Vdc
-13 Vdc	+0.818 Vdc ±0.025 Vdc

TABLE III. Vertical earth's rate compensation.

Heading	MAG VAR	Roll Output	Pitch Output
0°	0°	1.27 Vac ±0.1V	0 Vac ±0.05V
345°	+15°		0 Vac ±0.05V
15°	-15°		0 Vac ±0.05V
90°	0°	0 Vac ±0.05V	1.27 Vac ±0.1V

TABLE IV Crash safety test levels.

	Peak Amplitude (g units)	Nominal Duration (Milliseconds)
Service shock (equipment normally mounted)	15	11
Service shock (equipment removed from resilient shelf or panel)	15	22
Crash safety (equipment normally mounted)	30	11
Crash safety (equipment removed from resilient shelf panel)	30	22

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:54Z
Check the source to verify that this is the currentversion before use.

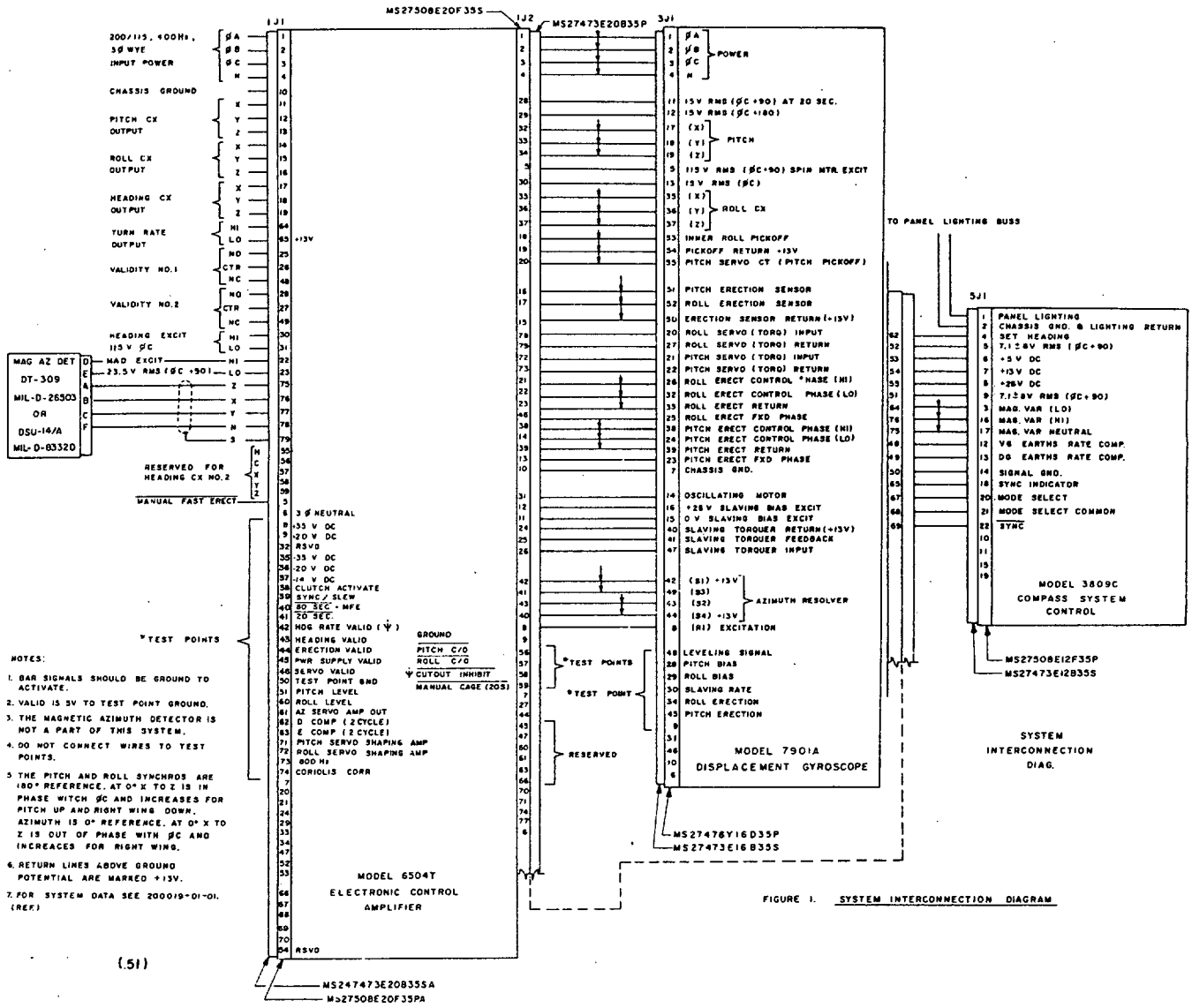


FIGURE 1. SYSTEM INTERCONNECTION DIAGRAM

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

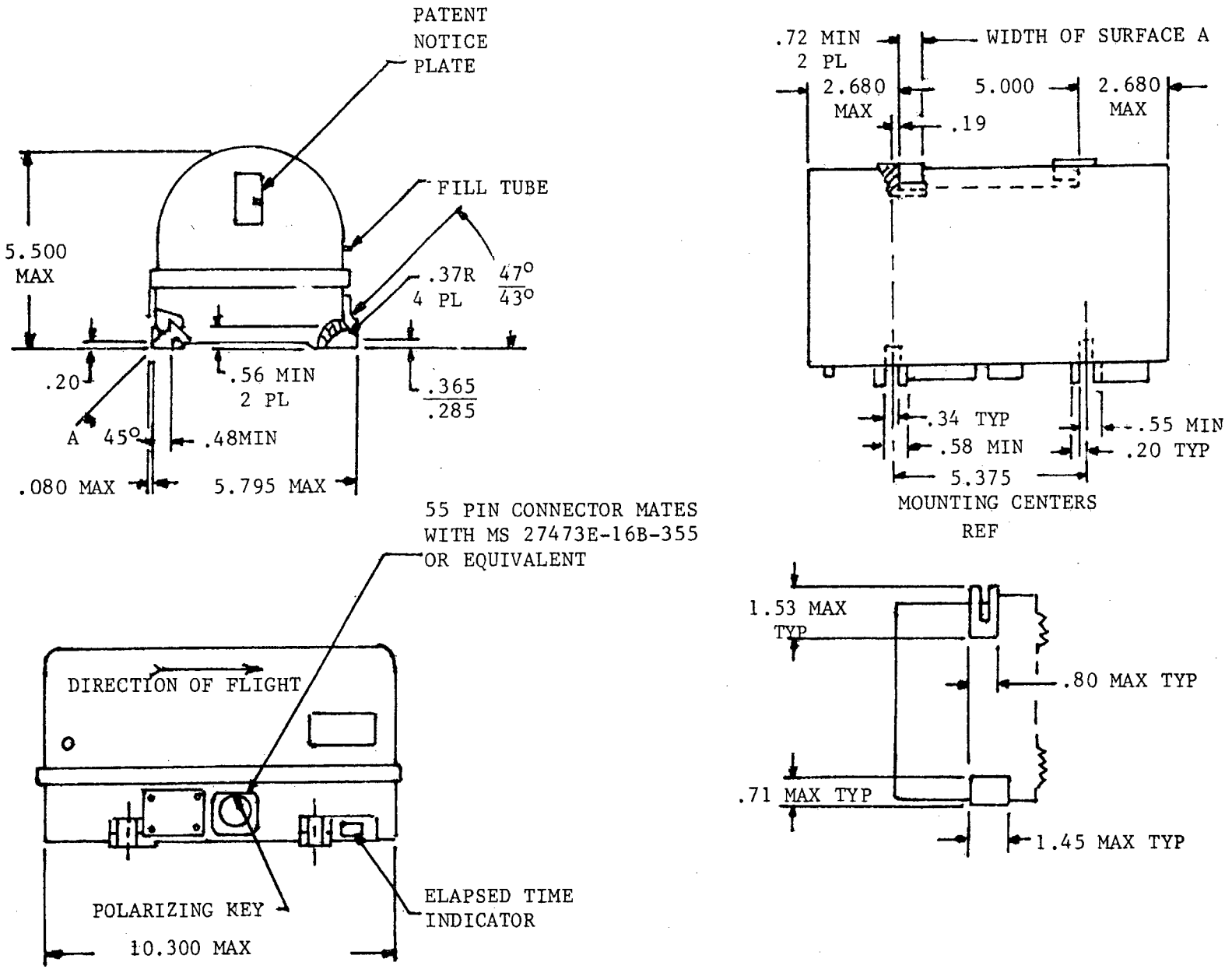


FIGURE 2. Displacement Gyroscope (DG).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

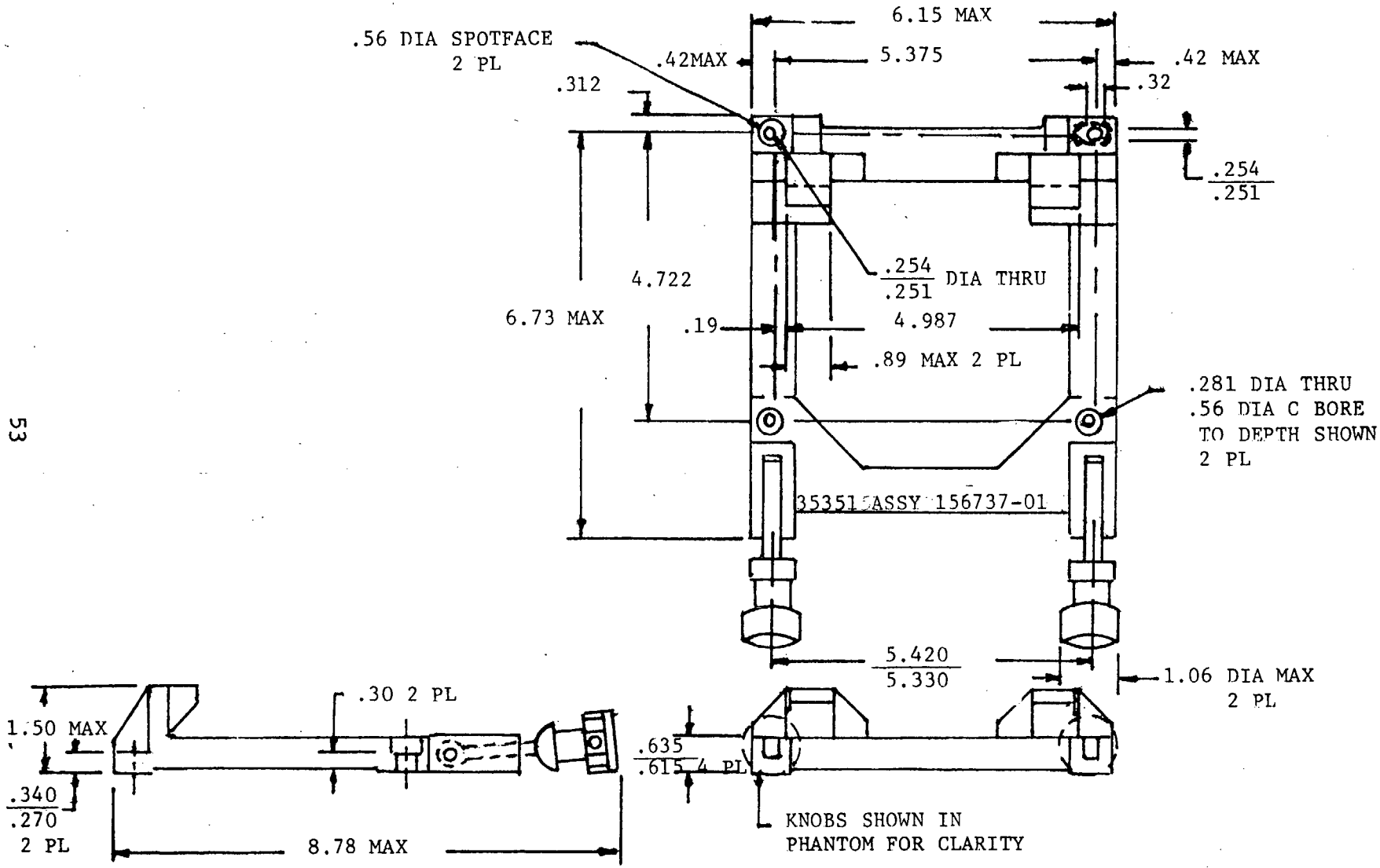


FIGURE 3. D G mounting base.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

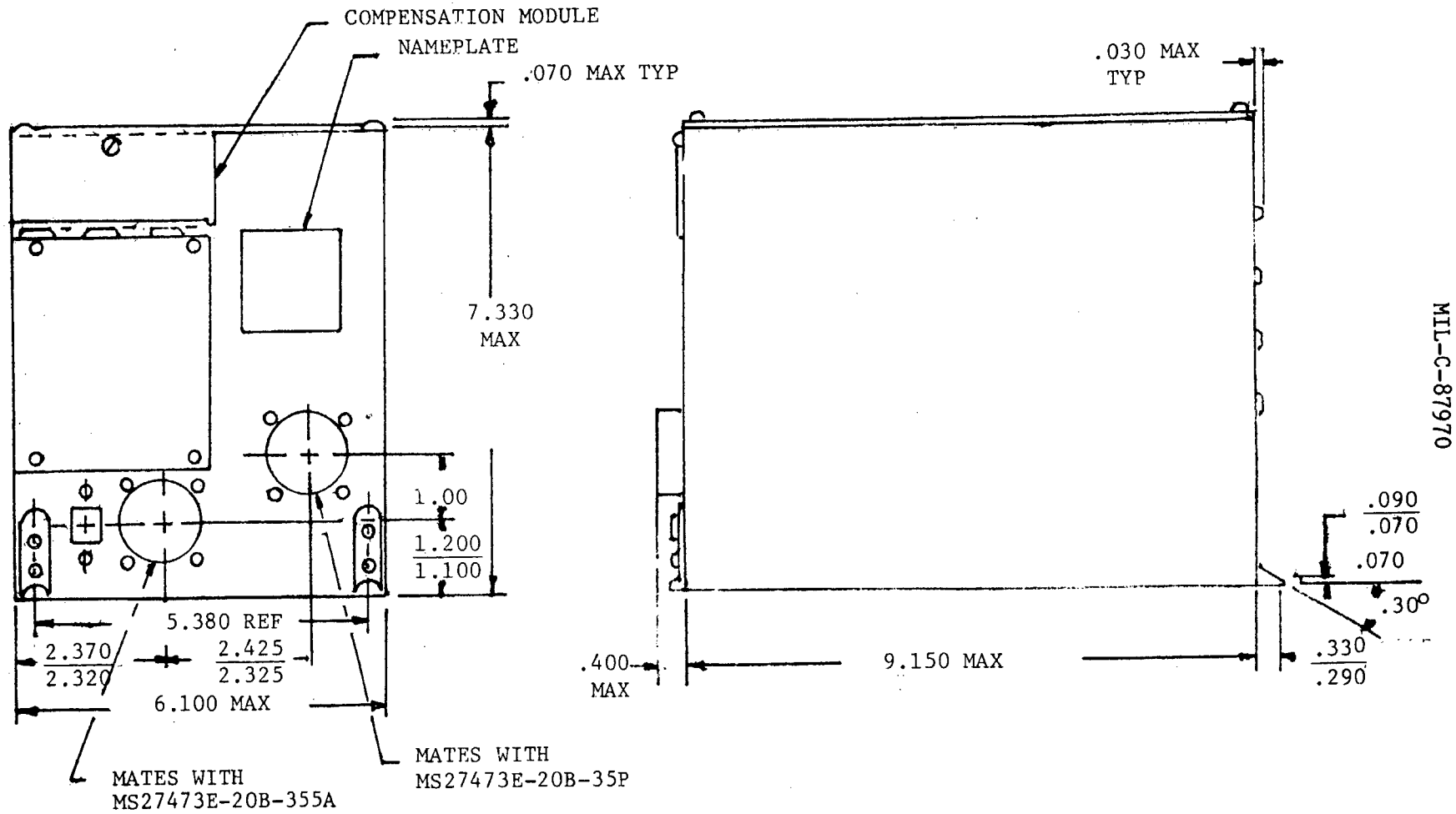


FIGURE 4. Electronic Control Amplifier (ECA).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

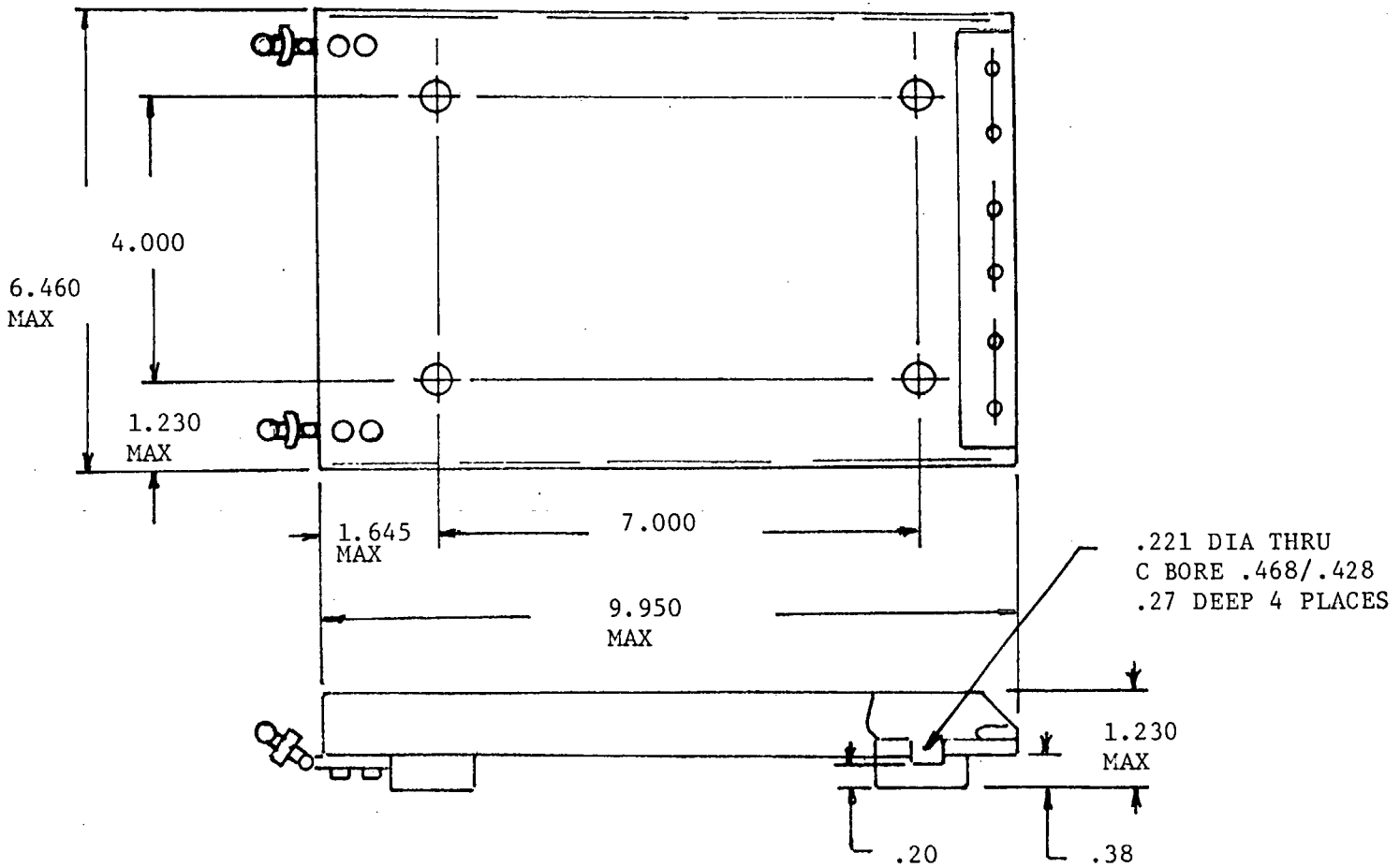


FIGURE 5. ECA mounting base.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

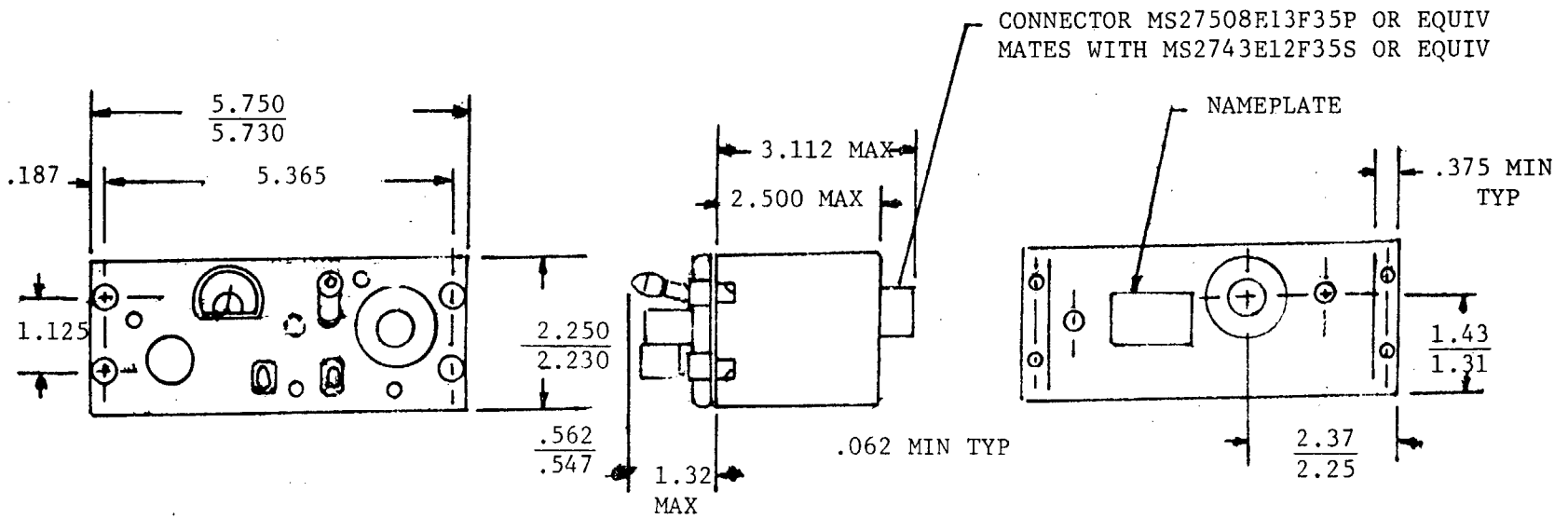


FIGURE 6. Compass System Controller.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

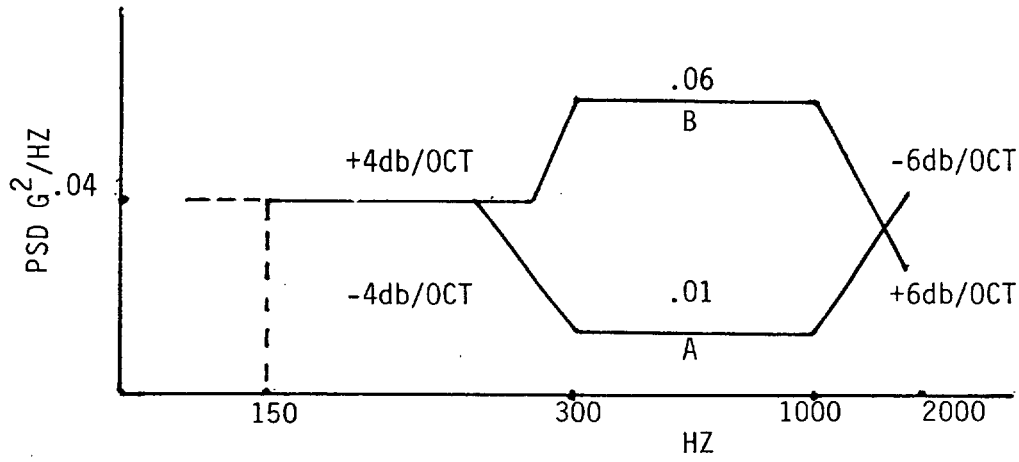


FIGURE 7. Random vibration - normal spectra.

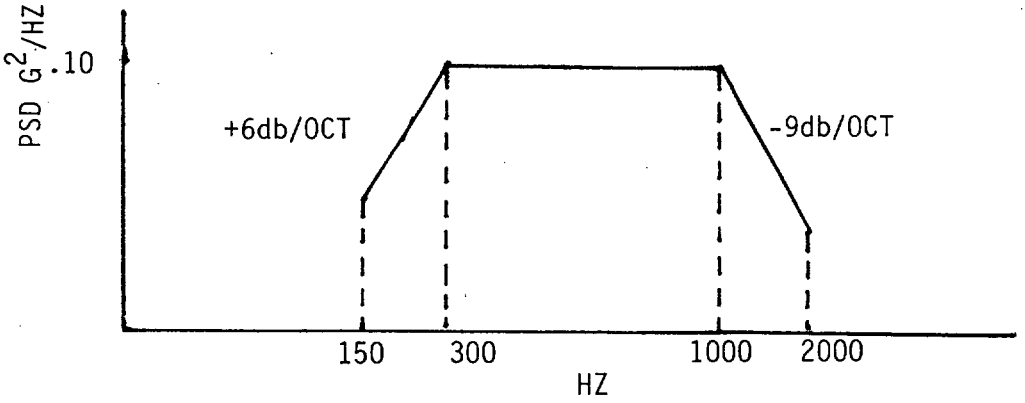


FIGURE 8. Random vibration - gunfire spectrum.

Custodian:
Air Force - 99

Preparing Activity:
Air Force - 71

Project Number:
6615-F058

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE AIR FORCE

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

OKLAHOMA CITY AIR LOGISTICS CENTER
ENGINEERING DIVISION
OPERATIONS AND SUPPORT BRANCH
OC-ALC/MMEDO
TINKER AFB, OK 73145

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.